

Call for Pilots Manual: Work Strand 2 - Creating new Local Digital Twins based on common needs

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Abbreviations and acronyms

Abbr.	Description	Abbr.	Description
AI	Artificial Intelligence	GA	Grant Agreement
ALTAI	AI HLEG's trustworthy AI assessment list	GDPR	General Data Protection Regulation
CA	Consortium Agreement	IDSA	International Data Spaces Association
Cerema	Centre d'études et d'expertise sur les risques, l'environnement, la mobilité et l'aménagement (Centre for Studies and Expertise on Risks, the Environment, Mobility and Urban Planning)	IEC	International Electrotechnical Commission
CfP	Call for Pilots	ISO	International Organization for Standardization
CFS	Certificate on Financial Statements	KPI	Key Performance Indicator
DA	Data Act	LDT	Local Digital Twin
DEP	Digital Europe Programme	LDT CitiVERSE EDIC	CitiVERSE European Digital Infrastructure Consortium
DGA	Data Governance Act	LDT4SSC	Local Digital Twins for Smart Sustainable Communities project
DS4SSCC	Data Space for Smart and Sustainable Cities and Communities project (preparatory action)	LIST	Luxembourg Institute of Science and Technology
DS4SSCC-DEP	Deployment project for the European data space for smart communities project (deployment action)	MIMs Plus	Minimal Interoperability Mechanisms
DSSC	Data Spaces Support Centre	OASC	Open & Agile Smart Cities
EC	European Commission	OCD	Ownership Control Declaration

ECA	European Court of Auditors	OCT	Overseas Countries and Territories
EDIC	EU Digital Infrastructure Consortium	OLAF	European Anti-Fraud Office
EEA	European Economic Area	PDF	Portable Document Format
EIF	European Interoperability Framework	PIC	Participant Identification Code
ENoLL	European Network of Living Labs	SIMPL	EU programme for Smart Middleware Platform, comprising of SIMPL Open, SIMPL Labs and SIMPL Live
EU	European Union	SME	Small and medium-sized enterprise
FAQ	Frequently Asked Questions	TalTech	Tallinn University of Technology
Fiware	FIWARE Foundation	TEF	Testing and Experimentation Facility (CitCom.ai)
FRAIA	Fundamental Rights and Algorithm Impact Assessment	TGB	Technopolis Consulting Group Belgium

Glossary

Term	Description
LDT4SSC Consortium	The LDT4SSC consortium includes a total of 10 partners: OASC, FIWARE, TalTech, ENoLL, Technopolis Consulting Group Belgium (TGB), Kereval (KEREVAL), The Luxembourg Institute of Science and Technology (LIST), Cerema, Libelium (LIBELIUM LAB) and UGent-IDLab (UGent).
Eligibility Check Grid	The checklist is used by the LDT4SSC consortium to check Call for Pilots applications against a set of defined eligibility criteria (legal, financial, technical, ethical).
Ethical Board	The Ethical Board comprises a distinct group of experts that is established to assess ethical issues in the LDT4SSC Project. They will be asked to propose mitigation measures when ethical issues are identified.
Evaluation Committee	The Evaluation Committee covers a dedicated group of experts from LDT4SSC partners and external stakeholders who are responsible for evaluating the Call for Pilots applications. The intention is to have different Evaluation Committees nominated to evaluate the different Call for Pilots which happens across three waves
Evaluation Team	The Evaluation Team is made up of selected people from the LDT4SSC project partners who have been assigned responsibility for organising the whole evaluation process and running the pre-screening of applications. Similarly, the Evaluation Team is tasked to provide both the final ranking and selection of the pilots.

Third-party	A third party is, by definition, anyone not directly involved in the grant: neither the contracting authority nor the beneficiary (including coordinator, co-beneficiaries if any or affiliated entities if any)
Lead Partner	A Lead partner represents the legal entity that is responsible for the pilot implementation and carries out the coordination of the concerned pilot. It is preferred that the lead partner is a local or regional public administration in the EU Member States.
Pilot Consortium	A Pilot Consortium refers to a collaboration of two or more participating organisations teaming up to jointly implement a project or an activity within a project.
Steering Committee	The committee responsible for making decisions regarding third-party funding, based on the evaluations conducted by the Evaluation Committee(s) and the Ethical Board. It consists of one representative from each private entity in the LDT4SSC consortium.

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Executive summary

This manual serves as a comprehensive guide to the Call for Pilots (CfP) inviting applicants to submit pilot proposals under Work Strand (WS) 2 - Creating new Local Digital Twins based on common needs. This CfP is launched in the frame of the Local Digital Twins for Smart and Sustainable Communities (LDT4SSC) project funded by the European Union. The objective of the LDT4SSC project is to build a robust, scalable ecosystem of local digital twins and stimulate the market for innovative, Artificial Intelligence (AI)-driven Local Digital Twin (LDT) services. The project also seeks to address barriers to AI adoption in Europe, such as limited investment, regulatory complexity, and gaps in expertise. It supports the development of advanced AI-driven digital twin solutions by pooling public sector demand and ensuring a fair, competitive environment for innovators.

This document provides a comprehensive description of the overall CfP process, covering the scope of the call, its framework, the financial, ethical, and technical specifications, and the support to applicants. It further sets out the timeline, the evaluation, and the journey towards contract signing. This document also includes several Annexes covering all the information relevant to announcing the Call and supporting documentation for the CfP application.

This CfP is part of the open calls foreseen in the LDT4SSC project and will cover three WS. WS1: Interconnecting existing LDTs to create a federated European Union (EU) wide network supporting seamless data exchange; WS2: Creating new LDTs based on common urban challenges like mobility, energy, air-quality and waste management to foster replicable solutions; and WS3: Advanced AI-based tools and innovative open-source components to enhance LDT capabilities with immersive and predictive services. This is the second CfP and it is targeted at pilot applicants interested in applying under WS1, WS2 and WS3. The implementation of the pilots in each call is envisioned to last between 12 and 18 months.

The WS2 is aimed at the following stakeholders: Public administrations at local, regional and national levels, serving the needs of local communities, together with their partners such as businesses, technology developers, research institutions and academia, and non-governmental organisations. The total EU funding available for the pilots in this call amounts to at least €5.8 million. The following financial rules apply:

- The **maximum grant awarded per third-party across pilots** is **€500,000**;
- The **maximum cumulative grant per consortium** is **€1,000,000**;
- **Applicants** are required to **cover at least 50% of total pilot costs**.

This call for pilots manual is structured as follows:

- Chapter 1 provides the Executive Summary and Introduction to the project and open calls.
- Chapter 2 outlines the broader framework for the open call, by providing the scope for the Work Strand and the objectives. It further describes who can apply, the financial framework, the non-technical framework and the technical framework for the pilots.
- Chapter 3 covers the process and timeline for the open call as well as instructions on how to apply.
- Chapter 4 covers the evaluation framework and selection criteria.

- Chapter 5 explains the support infrastructure available for applicants.
- Chapter 6 and 7 provide an overview on training and supporting activities as well as the monitoring and impact framework.
- Lastly, Annex 1 includes the different documentation which forms part of the application package, covering the templates for the Financial Form, the Commitment Letter, the Ethics and Data Protection, the Letter of Intent and the Ownership Control Declaration. Annex 2 provides additional materials containing information for context regarding the LORDIMAS digital maturity assessment, the Local Digital Twin capabilities, LORDIMAS digital maturity assessment, Capabilities of a Local Digital Twin, European Green Deal Policy areas, New European Bauhaus initiative values and LDT4SSC key challenges, European Green Deal Policy areas, Example of a (twice!) cross-sectorial use case, Data governance, NACE (“statistical classification of economic activities” in the European Community), High-Level architecture Overview of an LDT with the EU LDT Toolbox, LDT4SSC Methodology, and a List of pilots projects produced assets.

1. Introduction

1.1 About the Local Digital Twins for Smart and Sustainable Communities

The LDT4SSC project is funded by the Digital Europe Programme and has the following key objectives:

1. **Connecting existing LDTs from cities and communities** to create a federation of LDTs across the EU. Reinforced interoperability through the aggregation of LDTs at a larger scale (cross-sectors, cross-cities and cross-borders) will help in scaling up European common data sets and open-source solutions. It will also facilitate less advanced cities and communities joining the existing EU LDT ecosystem.
2. **Expanding existing local LDTs with new open-source LDT services based on shared needs of cities and communities.** These services should aim to improve decision-making processes and citizen interaction, reduce risks, costs, downtime and enhance resilience and sustainability of LDT platforms while enabling new value creation.
3. **Complementing the EU LDT Toolbox launched under the Digital Europe Work Programme 2021-22 with additional AI-based and innovative services** (e.g. for adaptable multi-sector considerations, advanced simulation and modelling approaches including bottom-up self-organised models). The AI services will be developed and tested within existing cities/communities and be replicable in other contexts.

These objectives form the foundation of an LDT ecosystem that empowers communities to collaborate effectively while advancing Europe’s digital transformation.

This project seeks to build a robust, scalable network of Local Digital Twins (LDTs) and stimulate the use of innovative, data-driven and AI-driven LDT services. It builds on- and integrates- outputs from related EU-funded projects and initiatives, such as the deployment

project for the European data space for smart communities (DS4SSCC-DEP)¹, the EU LDT Toolbox², CitiVerse EDIC projects³ Living-in.EU⁴, CitCom.ai⁵, and the Smart Middleware Platform (SIMPL)⁶ leveraging existing knowledge, tools, and stakeholder networks. By doing so, this project contributes directly to the Digital Europe Programme's (DEP) objectives of enhancing technological sovereignty, promoting ethical AI, and fostering an inclusive digital market, while integrating ongoing initiatives including the Networked Local Digital Twins towards the CitiVERSE European Digital Infrastructure Consortium (LDT CitiVERSE EDIC), as well as supporting policy goals like the New European Bauhaus⁷, the Green Deal⁸, and the Digital Decade⁹. The figure below shows how LDT4SSC is positioned in the wider network of initiatives and the evolution from the Data Space for Smart and Sustainable Cities and Communities project (DS4SSCC) to LDT4SSC. Together, these efforts will establish a resilient LDT infrastructure that enables European communities to optimise urban services, share resources, and tackle shared challenges such as climate change, air quality, waste, and energy efficiency.

¹ The European data space for smart communities is an EU-wide action creating a cross-sectorial data space for governments on all levels and their providers to deliver the best possible services to their citizens by enabling interoperability to reach critical goals: <https://www.ds4sscc.eu/>

² The EU Local Digital Twin (LDT) Toolbox is a flagship initiative of the European Commission that provides a modular, standards-based suite of tools designed to help cities and communities across Europe simulate, analyse, and plan urban environments more effectively: <https://interoperable-europe.ec.europa.eu/collection/ldttoolbox>

³ <https://digital-strategy.ec.europa.eu/en/factpages/citiverse>

⁴ Living-in.EU is an EU initiative for local and regional leaders who believe that technology can help them make their town, city, or region a better place to live: <https://living-in.eu/>

⁵ Testing AI in Smart Cities and Communities: <https://citcomtef.eu/>

⁶ SIMPL is an open source, secure middleware that supports data access and interoperability in European data initiatives: <https://simpl-programme.ec.europa.eu/>

⁷ New European Bauhaus (NEB) is a policy and funding initiative that makes green transition in built environments and beyond enjoyable, attractive and convenient for all: https://new-european-bauhaus.europa.eu/index_en

⁸ The European Green Deal is a set of policy initiatives by the European Commission with the overarching aim of making the European Union (EU) climate neutral in 2050: https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en

⁹ The Digital Decade policy programme, with concrete targets and objectives for 2030, guides Europe's digital transformation: https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age/europes-digital-decade-digital-targets-2030_en

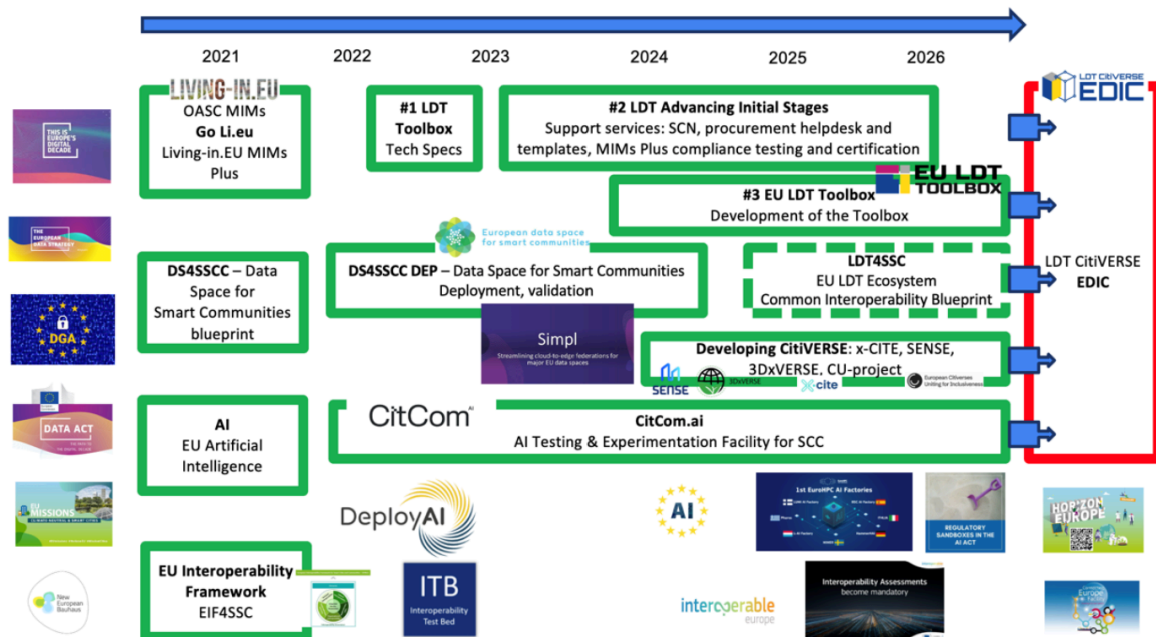


Figure 1. From DS4SSCC to LDT4SSC: LDT4SSC Smart Communities Ecosystem

The project also addresses broader barriers to AI adoption in Europe, which include limited investment, regulatory complexity, and gaps in expertise. By pooling public sector demand and ensuring a fair, competitive environment for innovators, it supports the development of advanced AI-driven digital twin solutions. This approach benefits both the public and private sectors, aligns with the European Union EU priorities for ethical AI and open standards, and reduces reliance on foreign technologies.

The LDT4SSC project is implemented by 10 partners, outlined in Table 1.

Table 1. LDT4SSC partners

In close alignment with the LDT CitiVERSE EDIC, the project accelerates the deployment of interoperable digital infrastructures across Member States, fostering a cross-border collaboration. In doing so, it contributes to the DEP's vision of a resilient, unified digital landscape.

2. Call for pilots framework

2.1 Aim of the calls for pilots

The aim of the open calls for pilots is to engage European local and regional authorities and its various stakeholders, ranging from public administrations at local, regional and national levels, businesses, technology developers, suppliers, research institutions, and academia to participate in building a networked ecosystem of LDTs. The calls are designed to ensure transparency, equal treatment, and compliance with EU standards, including legal, financial, ethical, and technical requirements.

The Open Calls are structured around **three specific Work Strands (WS)**, outlined below, each targeting a key aspect of the LDT ecosystem.

Ultimately, the calls aim to select high-quality pilot projects that can contribute to creating an interconnected LDT ecosystem, promote collaboration among stakeholders, and accelerate the deployment and adoption of advanced digital twin services across Europe.

Work Strand 1 (WS1) focuses on interconnecting existing LDTs to create a federated EU-wide network supporting seamless data exchange and interoperability;

Work Strand 2 (WS2) aims at creating new LDTs based on common urban challenges like mobility, energy, and sustainability to foster replicable solutions;

Work Strand 3 (WS3) develops new advanced AI-based capabilities to the LDTs Toolbox. Integrating AI-driven, added value services to enhance existing LDT.

Together, these strands support scalable, sustainable, and inclusive digital transformation aligned with European policies and market development.

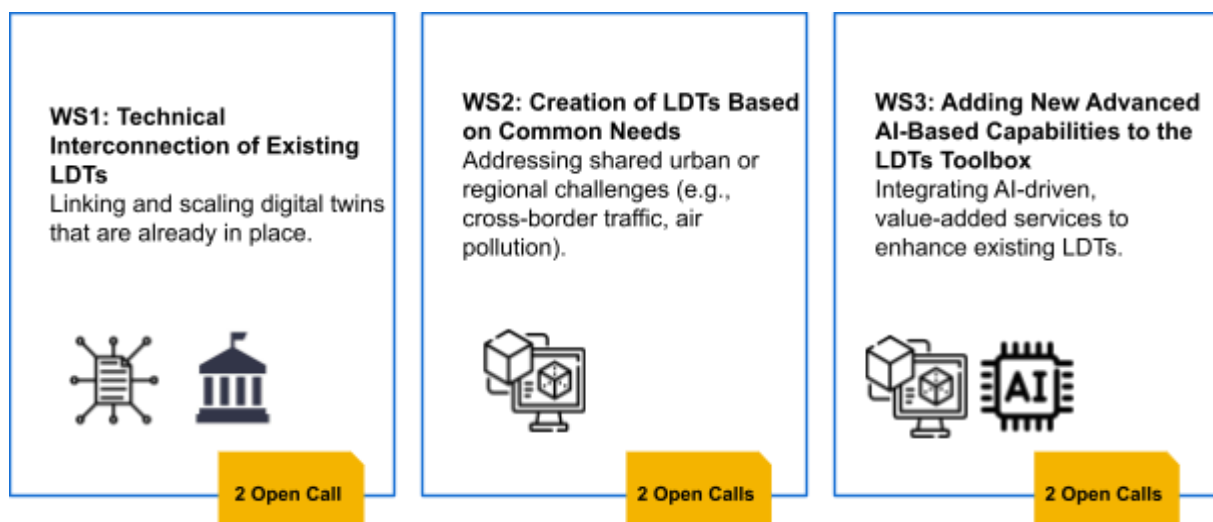


Figure 2. LDT4SSC three Work Strands across five Open Calls

This document focuses on the technical specifications for Work Strand 2, explained in the following section.

2.2 Scope and targets of Work Strand 2

WS2 seeks to create new Local Digital Twins based on common needs. The intent of the WS2 pilot activities is to bring together groups of cities and communities and partner organisations around concrete challenges that occur across cities and communities (e.g. transboundary mobility corridors, sectoral public services that multiple municipalities must manage in coordination, etc.) and **co-design services** to tackle these.

Through WS2 pilots, participating cities and communities will produce operational LDT services and packaged assets that demonstrate how shared data, common data models and reusable software components can be combined to deliver measurable improvements in the community. Hence, WS2 pursues the deployment of full-scope, user-driven LDTs with advanced capabilities that create tangible value due to the co-designed services for public authorities, citizens and local stakeholders. Pilots will serve as living examples of replicability: the outputs should be conceived to be transferable to other contexts, and documented and prepared for reuse so that results scale across regions and are replicable across the EU.

WS2 also invites technology providers willing to improve their value propositions and accelerate the adoption of their solutions in European and global markets by aligning with public sector needs and priorities. The development of LDTs around common challenges strengthens the competitive edge of technology providers by creating a more focused market for solutions that directly address high-demand policy areas.

The produced assets will populate the LDT4SSC Service and Assets Repository, which should also be integrated to the LDT Toolbox Marketplace and the DOME with industry-specific, advanced services.

A central ambition of WS2 is to demonstrate how newly created LDTs can federate with existing LDT networks established in WS1. By pursuing federation, supported by the LDT4SSC consortium, WS2 pilots will show how independently developed LDTs can interconnect using SIMPL, exchange data and services and leverage data spaces to address multi-community challenges at scale. This federated approach enables pilots to combine strengths: WS1 pilots that focus on interlinking mature LDTs provide operational pathways and interoperability lessons, while WS2 pilots expand the ecosystem by introducing newly developed LDT instances with services tailored to common needs. The interaction between WS1 and WS2 thus fosters an emergent, networked LDT marketplace in which services developed for one set of communities are discoverable, described and reusable by others.

2.3 Consortia eligibility

The Open Call aims to address a diverse variety of local and regional communities. Accordingly, participation is open, but not limited, to:

- Public administrations at local and regional levels
- Public administrations at national level, European Digital Infrastructure Consortia (EDICs) such as the LDT CitiVERSE EDIC
- Businesses, technology developers, and suppliers
- Research institutions and academia
- Non-governmental organizations (NGOs), non-profit organisations and other civil society actors

Each pilot consortium must comprise **at least two local and regional authorities** drawn from two different eligible countries. Eligible countries include:

- EU Member States, including their outermost regions, (together with their overseas countries and territories (OCTs))
- Non-EU countries (Norway, Iceland, Liechtenstein)
- Listed European Economic Area (EEA) countries and DEP Associated countries¹⁰

Additionally, in order to ensure replicability and scalability potential of the pilot, each consortia is required to involve at least **one additional partner** drawn from one of the following categories:

- Private entity (e.g., service provider)
- Private association (with legal status)
- Trusted third party
- Private representative of a use-case sector

Member(s) of the LDT4SSC Consortium **are not allowed** to be part of the pilot consortia.

Owing to security considerations, **restrictions apply to the composition of pilot consortia:**

- Entities must not be under the control of an ineligible country. Where there is a risk of such control, the entity will be required to provide a guarantee, which will be subject to assessment.
- All participants, with the exception of entities already validated as public bodies by EU Member States, must complete an **Ownership Control Declaration¹¹ (OCD)**. This must be self-declared at the proposal stage, and, in the case of a successful award, the official OCD must be submitted within 14 days of notification.
- Should an assessment reveal that a consortium member is controlled by an ineligible country, participation will be conditional on the submission and acceptance of an adequate guarantee. These guarantees will be examined on a case-by-case basis, taking account of the specific legal and factual circumstances.
- Participation is restricted to entities established in eligible countries, whether acting as beneficiaries, affiliated entities, associated partners, subcontractors, or recipients of financial support to third parties.
- All project activities, including subcontracted work, must also take place within eligible countries.
- Participants from DEP-associated countries (with the exception of EEA members) are required to provide an appropriate guarantee, approved by their country of establishment, in order to comply with the provisions of this Call for Pilots Manual.
- While consortia may, in principle, be led by any eligible partner, preference is given to **EU local and regional authorities as Lead Partners**. Where this is not the case, additional eligibility checks will be applied, covering ownership control, solvency, and state aid compliance. These checks may delay the commencement of the pilot or affect pre-financing arrangements.

¹⁰

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/digital/guidance/list-3rd-country-participation_digital_en.pdf

¹¹https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/common/temp-form/af/ownership-control-declaration_en.docx

- The grant agreement (GA) will be signed exclusively with the Lead Partner, who will hold primary responsibility for project implementation. Consortium partners are, in turn, required to conclude a Consortium Agreement (CA) among themselves.

Applicants should further note that **no organisation (legal entity) may be awarded more than one grant as Lead Partner under this framework**. Where multiple applications from the same organisation are shortlisted in the respective Open Calls, only the highest-ranked proposal will be retained for funding.

The organisation's role as a Lead Partner will be checked in each final ranking process to determine whether it had already been awarded funding in previous calls. For this purpose, public administrations or universities will be treated as a single organisation represented by one legal entity, irrespective of the functional independence of their departments or units.

Final checks during the ranking phase will verify participation using, inter alia, the organisation's national registration number (e.g. VAT, national code, social security number, Participant Identification Code (PIC)), its legal name, and the name of its legal representative.

Organisations (legal entities) can be involved in multiple grants under this framework, however, organisations can at most be acting as Lead Partner in one pilot throughout the entire open calls.

2.4 Financial framework

2.4.1 Form of grant, maximum grant amount and reimbursement rate

Financial support will be provided in the form of grants, reimbursed on the basis of actual eligible costs incurred and reported through interim and final reporting. **Indirect costs** will be calculated as a **flat rate of 7% of eligible direct costs**, in line with the rules of the DEP.

The total EU funding available for the pilots amounts to EUR17 million (85% of the total budget of the LDT4SSC project). Under this CfP, the LDT4SSC project will allocate €5.8 million to Work Strand 2.

The following financial rules apply:

- The **maximum grant awarded per third-party across pilots is €500,000 (across multiple consortia and awarded pilot projects)**. Any expenses exceeding the maximum grant must be covered entirely by the pilot consortium members through their own resources;
- The **maximum cumulative grant per consortium is €1,000,000**;
- **Applicants** are required to **cover at least 50% of total pilot costs**.

Consortia applying to the CfP must provide a detailed budget, presented in **euros**. Applicants based in non-EU countries are required to apply the conversion rates published in the C

Series of the Official Journal of the European Union¹². Responsibility for any exchange rate risk lies entirely with the applicants; further provisions will be outlined in the GA.

The budget must be properly balanced, with total income (requested EU contribution plus own co-financing) matching total expenditure. Expenditure should refer exclusively to the estimated costs necessary for the implementation of the pilot.

2.4.2 Co-funding model

The EU contribution for WS2 amounts to at least €5.8 million, to be distributed across approximately 5-7 selected consortia.

The co-funding model ensures that **EU support covers up to half of the eligible expenses**. **Applicants** demonstrate their capacity to **sustain at least 50%** of the pilot costs through their own resources as direct co-financing. This approach guarantees shared investment and responsibility between the EC and the participants, fostering stronger commitment and enhancing the sustainability of the pilot actions. Applicants are therefore expected to **clearly outline their funding strategy** when submitting proposals.

The co-financing must be a dedicated, auditable cost, explicitly indicated in the pilot budget and documented in accordance with normal accounting practices. Only direct monetary contributions or monetised resources included in the budget are eligible. In-kind contributions not monetised in the budget are not admissible.

All costs must be **actual, incurred during the pilot period**, and in full compliance with applicable regulations, including state aid rules and the prevention of double funding.

Participating local and regional administrations as well as any other member of the consortium contributing financially to the pilot activities are required to demonstrate their capacity to provide at least 50% of co-financing with a **Letter of Commitment**¹³.

2.4.3 Eligibility of costs

Eligible direct costs must be clearly attributable to the pilot and directly linked to its implementation. To qualify, costs must:

- Be incurred by the beneficiary during the pilot period, excluding costs related to final reports and audit certificates;
- Be included in the estimated overall budget attached to the GA¹⁴;
- Be necessary for the execution of the pilot;
- Be identifiable and verifiable, recorded in the beneficiary's accounting system in accordance with national accounting standards and the beneficiary's usual cost-accounting practices;
- Comply with applicable tax and social legislation;
- Be reasonable, justified, and adhere to principles of sound financial management, including economy and efficiency.

Additional considerations include:

¹² ECB website

https://www.ecb.europa.eu/stats/policy_and_exchange_rates/euro_reference_exchange_rates/html/index.en.html

¹³ Letter of Commitment can be found on the Project Website at <https://ldt4ssc.eu/call-one/>

¹⁴ Financial Form Template can be found on the Project Website at <https://ldt4ssc.eu/call-one/>.

- Value-added tax (VAT) is eligible where it is non-recoverable under national legislation and paid by a taxable person, in line with Article 13(1) of Council Directive 2006/112/EC;
- Costs must be necessary and proportionate, avoiding excessive or unnecessary expenditure;
- Accounting and auditing procedures must allow reconciliation between declared costs and supporting documentation;
- Documentation justifying costs must be retained for five years following the closure of the project.

According to DEP rules¹⁵, eligible direct costs comprise **Personnel, Subcontracting, and Purchase Costs**, while indirect costs are eligible at a flat rate of 7% of total direct costs.

Cost category	
A. Personnel costs	
B. Subcontracting costs	
C. Purchase costs	C.1 Travel and Subsistence
	C.2 Equipment
	C.3 Other Goods, Works, and Services
D. Indirect costs ($=7\% \times (A+B+C)$)	
Total eligible costs (A+B+C+D)	

Personnel costs

This category covers salaries and associated costs (social security, taxes, other statutory contributions) for personnel employed by the beneficiary, including seconded staff. Time spent on the pilot must be documented through a time-recording system (e.g., timesheets).

Subcontracting costs

Subcontracting is permitted only for specific action tasks defined in the pilot and must be included in the estimated budget. Subcontracts must follow competitive procedures to ensure best value for money and avoid conflicts of interest. Subcontracting may not cover pilot management, coordination, or tasks performed by consortium members.

Purchase costs

This category includes costs for goods, works, or services necessary for pilot implementation, subdivided as follows:

C.1 Travel and Subsistence – Eligible travel and subsistence costs for personnel or explicitly budgeted experts related to pilot activities.

¹⁵ For comprehensive information on the eligibility of costs, please consult the [Annotated Grant Agreement EU Grants](#) and the [DEP MGA](#), particularly Art.6.

C.2 Equipment – Depreciation or rental costs of equipment used exclusively for the pilot, calculated according to national or institutional accounting practices.

C.3 Other Goods, Works, and Services – Additional costs such as dissemination, communication, and audit certificates.

Indirect costs

Indirect costs, which cannot be attributed directly to the pilot but are necessary for its administration, are limited to a flat rate of 7% of total direct costs (Personnel + Purchase + Subcontracting). This may include overheads such as utilities and office expenses. No supporting documentation is required for indirect costs.

Ineligible costs

Costs are not eligible if they:

- Represent capital returns, debt service, provisions for future losses, interest, doubtful debts, or currency exchange losses;
- Constitute excessive or reckless expenditure;
- Include deductible VAT;
- Are incurred after suspension of the pilot.

Additionally, the following are out of scope: infrastructure costs, sub-grants or prizes, and large research infrastructure.

2.4.4 Reporting

The Lead Partner of the Consortium is responsible for submitting both the interim and final reports on behalf of the consortium. The interim report should be prepared at the halfway point of the pilot's duration, while the final report should be submitted within 30 days of the pilot's conclusion.

The **interim reporting package** must include:

- A **Technical Report** outlining the activities carried out by consortium members, the use of resources, and updates on key performance indicators (KPIs), supported by evidence demonstrating tangible changes from the initial situation towards the expected impacts, as well as updates on risks.
- A **Financial Report providing a consolidated overview for the entire consortium**, including detailed information from all partners, documenting eligible costs such as actual direct costs and the 7% flat-rate for indirect costs.

The **final reporting package** must include:

- A **Technical Report** summarising the work carried out and the use of resources;
- A **Financial Report** containing individual financial statements from each member, similarly documenting eligible costs. All entities requesting contributions of **≥ EUR 325,000** must submit a **Certificate on Financial Statements (CFS)**. For further details regarding the CFS, please refer to [Section 2.5.6](#).

Pilot consortia are required to comply with contractual obligations under the Grant Agreement (GA), including provisions on Conflict of Interest (Article 12), Confidentiality and Security (Article 13), Ethics (Article 14), Visibility (Article 17.2), Rules for Implementation (Article 18), Information (Article 19), and Record-keeping (Article 20). Further details are provided in the Annotated Grant Agreement¹⁶.

Pilot Deliverables

Applicants are expected to submit a workplan revolving around the LDT4SSC **pre-defined Pilot Deliverables** (PD). No additional deliverables are required.

Nr.	PD Title	PD scope	Expected due date
PD1	Data Management Plan & Ethics	Defining data handling, contracting, storage, sharing and preservation practices, together with the ethical principles and compliance measures governing all data use.	M1-2
PD2	Interoperability Self-assessment	Complete the LDT4SSC Interoperability Self-Assessment tool at the start and in the second reporting period of the project. A score above three at the end is a condition for payment.	1. M1-2 2. second reporting period
PD3	Pilot Scope & Architecture	Defines the pilot by outlining its priority use cases, key stakeholders, and functional requirements, and sets out the high-level system architecture. This includes identifying technical requirements and specifications, data sources and models, relevant standards, and the component framework necessary to develop an interoperable LDT and services aligned with the LDT4SSC blueprint. In the case of use of proprietary components, applicants are expected to provide a clear development plan describing how the solution can be deployed using purely open-source technology.	<i>Provide due date</i>
PD4	Governance Scheme	Operating and legal conditions and contractual framework for the LDT and sharing of data and data models, including legal management of interoperability. Terms and conditions for sustainability, replication, scalability and sharing (legal clauses, contracts used, etc.).	<i>Provide due date</i>
PD5	Communication, Dissemination	Reports on the engagement strategy by detailing stakeholder engagement activities alongside communication, dissemination and	<i>Provide due date</i>

¹⁶ Annotated Grant Agreement. [Link](#).

Nr.	PD Title	PD scope	Expected due date
	on, Sustainability and Exploitation Report	marketing actions to support project visibility and involvement and promoting the services and their replicability. Provide a cost benefit analysis across social, environmental, organisational, and economic dimensions to substantiate your exploitation strategy, demonstrating that impacts have been assessed and measured (including economic impact as well as social and environmental impacts). Report on the updated plan for sustaining the LDT beyond project lifetime, with the designed business model, referencing also the recommendations from the Strategy for Sustainability and Growth from the LDT4SSC project.	

2.4.5 Payment procedures

Payment of the grant will be made in three instalments by Technopolis Consulting Group Belgium (TGB), the Party responsible for the Funding Support) to the Pilot Lead only, who will be solely responsible for distributing the grant among the consortium, as follows:

- **Pre-financing (30%):** Paid within 30 days of the GA signature to provide the consortium with initial resources to start project activities.
- **Interim payment:** Following the first reporting period, an interim payment may be made subject to a satisfactory interim evaluation. This payment will only be made **if the eligible costs reported at that stage exceed the amount of pre-financing already received.**

The total disbursed at this stage (**pre-financing plus interim**) **will not exceed 85% of the total eligible costs.** Payment will be made within 30 days of a positive evaluation and after the Party responsible for the Funding Support has received the corresponding funds.

- **Final payment (remaining balance):** payment of the remaining eligible costs, paid after successful completion of the pilot and approval of the final report, bringing total disbursement to 100% of the grant. Payment will be made after the Party responsible for the Funding Support (TechnoGB) has received the final payment from the Coordinator of the LDT4SSC project.

The financing remains the property of the Party responsible for the Funding Support (TGB) until the payment of the balance. In the case of any risk related to co-financing (50%) not being met during the grant awarding and/or any reporting stage(s), the consortium receives no pre-financing but gets reimbursed according to actual costs incurred at the end of the pilot, being certified by an external auditor.

2.4.6 Certificate of financial statement

In the event of an award, the Party responsible for the Funding Support (TGB), the EC, the European Anti-Fraud Office (OLAF), and the European Court of Auditors (ECA) reserve the right to conduct audits and checks on all aspects related to the grant, in accordance with Article 15 of the GA signed by the LDT4SSC project with the EC.

Awarded organisations requesting contributions of **≥ EUR 325,000** are required to submit a **Certificate on Financial Statements (CFS)** by an independent auditor with the final report. The CFS enables the granting authorities, TGB, OASC, EC, OLAF, and ECA, to verify that the costs declared in the financial statements are eligible. Please note that **this threshold is determined in accordance with EC regulations and may be subject to change should these regulations be revised.**

Costs incurred for producing the CFS are considered eligible and should be included under the **cost category C3, “Other goods, works and services”**. Applicants are therefore advised to account for CFS-related expenses in their pilot budget estimates

2.5 Work Strand 2 Framework

This section describes the technical and non-technical frameworks that define the LDT4SSC vision for the pilot project. These frameworks provide the **foundation for the requirements and recommendations** presented later (see ‘[Description of requirements & recommendations](#)’). Therefore, this section sets the framework within which requirements and operations operate, and pilots must therefore align with it to translate the project’s objectives and positioning into coherent pilot implementation.

2.5.1 Non-Technical Framework for Work Strand 2

The non-technical framework builds on existing and ongoing initiatives of the EU Smart communities ecosystem and is aligned and contributes to the European objectives regarding interoperability, digitalisation and the environment (Twin Transition, New European Bauhaus (NEB), Digital Decade, Green Deal, Data Strategy for a digital single market, Mission on Climate, Neutral and Smart Cities, Nature Restoration, Law, Soil Deal Mission, Restore our Ocean and Waters Mission). The overarching aim is to demonstrate **interoperable, replicable, scalable, and sustainable** digital solutions that can be replicated across Europe.

The approach to **interoperability in the LDT4SSC project follows a ‘by-design’ principle**, meaning that the ability of systems to exchange and use data with other systems is embedded from the outset rather than added later as an afterthought. In practical terms, this means that **technical, organisational, semantical and legal** choices are made from the outset to facilitate the smooth exchange of data and services with other systems or actors.

Solutions are expected to be designed for large-scale and cross-sector uptake, with clear pathways for scaling and integration into broader EU initiatives such as the LDT CitiVERSE EDIC¹⁷ and the LDT Toolbox Marketplace. This openness will enhance visibility, reduce

¹⁷ <https://digital-strategy.ec.europa.eu/en/policies/edic>

duplication, and accelerate adoption. The LDT4SSC project invites cross-border and cross-sectoral pilot initiatives to help implement and validate an interoperable ecosystem of LDTs in Europe. To enable replicability and scalability of pilot projects, these pilots will contribute to and operationalisation of the technical and non-technical LDT4SSC interoperability blueprint, which addresses: technical interoperability guidelines, **legal compliance and governance structures, viable business model strategies, value creation and economic sustainability, mechanisms for decision-making and coordination, transition strategies for future integration with the EDIC LDT CitiVERSE.**

Operationally speaking, this is made possible by pilots having methodological rigour. LDT4SSC proposes a development methodology with four stages: **(1) Explore, (2) Validate, (3) Define and (4) Implement across three thematic dimensions: governance, value creation and technical implementation.** The LDT4SSC consortium supports pilots in the operational implementation of this method with a set of activities and workshops. This methodology covers various topics, namely:

- **Define data governance** across its four dimensions -political, technical, legal and organisational- to facilitate the sustainability of the pilots project;
- **Have responsible digital practices** (ethics, eco-design of digital services, web-accessibility considerations, etc.) to create LDTs and services that are environmentally and socially responsible;
- **Perform a cost-benefit analysis** to get a comprehensive assessment of the digital solution indicating its financial as well as socio-environmental utility **relative to a baseline situation**;
- **Design a business model** to ensure the economic sustainability of the service after the pilot project;
- And **assess and measure impact** including economic impact as well as social and environmental impacts.

Following these steps and carrying out the proposed activities significantly increases the chances of success for the pilot project and sustainability.

Selected pilots will be part of a European-scale initiative aligned with the **EU Green Deal priorities**¹⁸ and **New European Bauhaus**¹⁹, targeting practical use cases addressing critical societal challenges through digitally-enabled innovation that demonstrate scalable, interoperable, and sustainable digital solutions replicable across Europe.

2.5.2 Technical framework for Work Strand 2

The technical framework for pilots in the LDT4SSC project also **builds on existing and ongoing initiatives of the EU Smart communities ecosystem**, also described in the project Knowledge Hub²⁰. This underlines the LDT4SSC ambition to **consolidate the existing architectures and best-practices into a clear and actionable Interoperability Blueprint** described in the previous sub-chapter. This blueprint will be updated to reflect the

¹⁸ For more information on the European Green Deal see https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en

¹⁹ To learn more about the New European Bauhaus initiative see https://new-european-bauhaus.europa.eu/index_en

²⁰ See <https://knowledgehub.ldt4ssc.eu/>

learnings of the pilots and can serve as a tool to facilitate and speed up the go-to-market for future digital twin developments.

From this prior work, the LDT4SSC project recognises that an **overarching meta-architecture is emerging for European LDTs and that it will allow a fair level-playing field for technology providers and procurers alike**. This meta-architecture, as shown in Figure 3 has interoperability at its core and will stimulate a strong and dynamic innovative market fueled by the public sector:

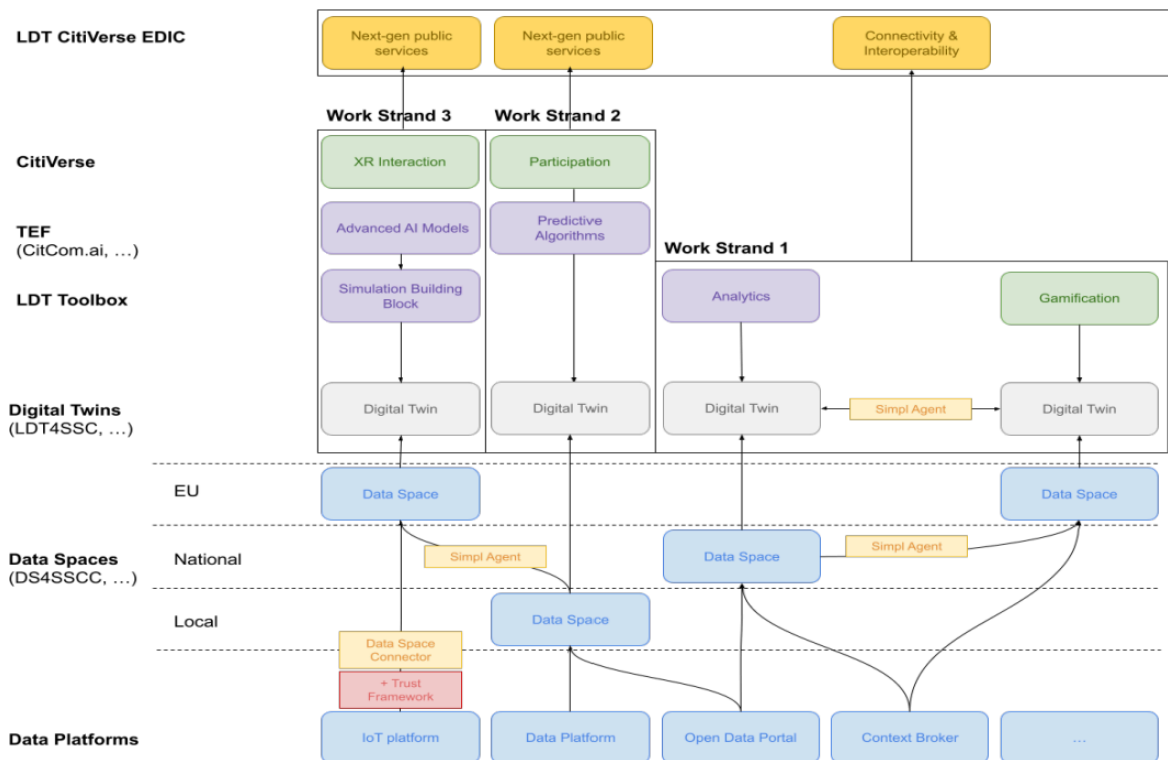


Figure 3. Meta-architecture for LDTs and Value-added services

1. **Data systems and Platforms that are set up within local ecosystems** are augmented with Trust Frameworks, Publication and Discovery Services, as well as Access and Usage Policies to make up a DS (DSSC Data Space Blueprint v2²¹, DS4SSCC-DEP Blueprint²²).
2. **Data Spaces (DSs) become interconnected**, through, among other mechanisms, Data Connectors (see the IDSA Data Connector report²³).
3. These local or regional DSs are expected to be **federated in European Data Spaces**, through, for example, the SIMPL framework²⁴, funded through DIGITAL Europe.

²¹ See: <https://dssc.eu/space/BVE2/1071251457/Data+Spaces+Blueprint+v2.0+-+Home>

²² See: <https://www.ds4sscc.eu/inventory>

²³ A comprehensive overview of available data connectors based on the IDS Reference Architecture Model: <https://internationaldataspaces.org/data-connector-report/>

²⁴ See footnote 5: <https://digital-strategy.ec.europa.eu/en/policies/simpl>

4. On top of these Data Spaces, **LDTs are developed using tools from the EU LDT Toolbox** and other components available through various suppliers.
5. Within the Testing and Experimentation Facilities (TEFs), **advanced AI and machine learning tools and models are made available** to further increase the effectiveness of the services developed within the LDTs.
6. **Interoperability is built into these value-added services**, thanks to the MIMs Plus²⁵ and the EIF²⁶. This way, these services can **benefit other communities**. They are therefore fed back into the LDT Toolbox Marketplace or the DOME and disseminated.
7. These **value-added services and the underlying connectivity, infrastructure and interoperability are planned to be sustained by the European Digital Infrastructure Consortium (EDIC)** –in this case, specifically, the LDT CitiVERSE EDIC (See 2.1 Non-Technical framework for further information on how operations are sustained).

2.6 Description of requirements & recommendations for pilots

2.6.1 Considerations

All listed requirements must be explicitly demonstrated in the application. Applicants are expected to demonstrate compliance primarily through the application form²⁷, which translates these requirements into clear questions, dedicated submission fields, and supporting guidance. The form is structured into four sections (*Pilot Details*, *Relevance*, *Implementation*, and *Impact*) each containing specific questions to guide applicants in describing their project. This will ensure that evaluators can verify compliance and compare proposals consistently. Note that these requirements do not cover administrative, financial, or ethical aspects, as those are addressed below in the CfP.

Requirements (Rq) are formulated across the following dimensions: *Pilot Details*, *Relevance*, *Implementation*, and *Impact*. Requirements are mandatory. Some are part of the pass/fail eligibility check and others are assessed by evaluators. Applicants must provide a convincing workplan (via the Application Form) describing how they intend to achieve the desired results.

²⁵ The minimal interoperability mechanisms (MIMs) Plus enable a minimal but sufficient I and f interoperability for data, systems, and services specifically in the context of smart city solutions: <https://living-in.eu/group/7/commitments/mims-plus-version-8-2025>

²⁶ See <https://interoperable-europe.ec.europa.eu/collection/iopeu-monitoring/european-interoperability-frame-work-detail> footnote 15:

²⁷ See Application Form at <https://ldt4ssc.eu>

Recommendations (Rc) are formulated across the same dimensions. They are optional but useful to improve the pilot's application, implementation, and potential to be sustainable.

2.6.2 Pilot Details

Rq1	Pilot applicants do not need to have an LDT in place to apply to this Open Call. However, see Rc1 for recommended levels of digital maturity.
Rq2	For each pilot member already operating an LDT or another data platform (GIS, open-data, data warehouse, data dashboards, etc.), provide: 1. A high-level description of the existing platform and its main capabilities. Specify which data sources are currently integrated and what services are offered within the city or community. 2. Where available, include a public reference (e.g. URL) to the platform; if not publicly accessible, provide representative screenshots of the platform's front-end. 3. A high-level description of the current architecture covering technical and functional aspects. Provide a diagram for each aspect (respectively a deployment diagram and an activity diagram). Specify the main tools, standards, and software components used, and illustrate the data lifecycle (from collection to use and sharing)
Rq3	Pilots must include at least one cross-sectoral²⁸ (See Annex 2.4 for an example and Annex 2.6 for NACE codes for sectors) use case that is innovative and citizen-focused with two services. • Cross-sectoral can mean: The developed services integrate datasets from two or more sectors (e.g. mobility, energy...); or • The LDTs provide data services that span at least two sectors within a single use case. Describe the use case, specifying the services to be delivered. Demonstrate that it is innovative and citizen-focused.
Rq4	Describe the current data governance (see explanation under Annex 2.5) scheme of each pilot site and the target data governance scheme of the pilot consortium across the political, technical, legal, and organisational dimensions.
Rc1	For each local authority/government involved in the pilot, please give the score obtained at the LORDIMAS digital maturity assessment²⁹ (see explanation in

²⁸ See list of sectors here: https://single-market-economy.ec.europa.eu/sectors_en

²⁹ [MaturityAssessment-help | LORDIMAS EN](#)

	Annex 2.1). The pilot lead is recommended to be at the ‘Digitally Optimised’ stage assessment, while all other participating authorities are recommended to be at least at the ‘Moderate’ stage in the LORDIMAS self-assessment upon application. All public authorities participating as pilot members are recommended to reach the ‘Digitally Optimised’ stage in the assessment, or to demonstrate further progress if they have already attained this level by the end of the project.
Rc2	To better demonstrate the replicability of the developed service, it is recommended that more than two public authorities join in pilot consortiums.

2.6.3 Relevance

Rq5	Public authorities part of the pilot consortium must share a common need and address a shared local challenge. The pilot consortium must provide, upon application, at least one letter of intent by another public authority (or service in another public authority if allowed), not it in the consortium, supporting that the local challenge addressed in the consortium is also shared in their city or community. Describe how the pilot addresses a shared local challenge between public authorities and how the developed LDT answers this common need. Precise what the ambition and rationale for jointly developing an LDT are.
Rq6	The use cases should align with the European Green Deal Objectives³⁰ or the New European Bauhaus³¹ priorities, or tackling key challenges identified by LDT4SSC (see explanation in Annex 2.3). Describe how the pilot addresses European priorities as well as the objectives of the LDT4SSC initiative and its ecosystem (for example, by advancing shared knowledge, tools, or methodologies).
Rq7	Each participating public authority must ensure both technical capacity (e.g., designated technical staff for implementation and continuity) and political support. Describe project management and coordination, including resource allocation. Provide an overview of the project teams, their collaboration approach, and any planned recruitment. If additional skills or resources are needed, explain how they will be secured through consortium partners, subcontractors, or other means, and specify the roles or profiles involved. Describe the political endorsement (see <i>Letter of Commitment</i>) of your pilot, specifying the type and extent of support. Where available, reference published or online materials (with links) or attach supporting documents that substantiate this endorsement .
Rq8	Pilots must actively engage end-users. Especially, end-users must test the implemented service before it is replicated. Specify the end-users of the planned services and describe how they will be engaged throughout the project lifecycle through workshops, consultations, or

³⁰ https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en

³¹ See https://new-european-bauhaus.europa.eu/index_en

	other participatory activities.
Rq9	Each pilot must involve at least three of the four Quadruple Helix stakeholder groups (public sector, private sector, academia/research, and civil society) in these engagement processes. List the involved and expected stakeholders across the four Quadruple Helix categories , outlining their roles, responsibilities, and engagement approach.
Rq10	Explain the broader applicability and relevance of your pilot . Indicate how its approach and/or results are relevant to be transferred to other EU communities.
Rq11	Describe how the project will align with and leverage the following relevant EU initiatives in their projects: guidelines and specifications (e.g. DSSC, Gaia-X...) and and/or deployment of tech stacks from other EU initiatives (SIMPL, LDT Toolbox, ...).

2.6.4 Implementation

Rq12	The project description and workplan should adopt the LDT4SSC methodology, following phases: Explore, Validate, Define and Implement, and cover the following key activities (See Annex 2.8). Applicants are expected to submit a workplan revolving around the LDT4SSC pre-defined Pilot Deliverables (PD) (See Pilot Deliverables).
Rq13	Pilots must develop a new common LDT based on their common need. Each public authority participating in the pilot must implement its own instance of the commonly developed LDT. Hence, there must be at least 2 instances of the common LDT, one for each public authority, which allows executing the two services related to the identified cross-sectoral use case. Each public authority must have access to at least one LDT. Accessible means that the LDT configuration is accessible through management interfaces for all 7 LDT Layers (See Annex 2.7). For instance: 1. Data Sources Layer: can add/change datasets and data sources; 2. Data Acquisition Layer: can configure data processing (ingestion and transformation); 3. Knowledge Layer: can deploy and parametrize different advanced analytics, algorithms and/or ML/AI models; 4. Interoperability Layer: can connect new interoperable systems and manage existing ones; 5. Services Layer: can add and administrate services and their interfaces; 6. Orchestration Layer: can connect other components (e.g. services, data models, simulations); 7. Visualisation Layer: can configure the way data is visualised.

Rq14	Describe how the pilot consortium will technically deploy their LDTs, specifying the tools, software, and standards selected for each component (e.g. data catalogue, management system, identity and authorisation mechanisms (IAM), data policies, connectors, and knowledge models). Highlight what is reused in the already existing infrastructures. Specify how the common LDT will be deployed (e.g. hosted solution, Software-as-a-Service (SaaS), on-premise) and explain how each participating public authority operates its own instance of the common LDT, how these instances enable the execution of the two services and how each authority will have management access to at least one LDT instance, including configuration and control through management interfaces covering all seven LDT layers. Applicants must complement the description with two draft diagrams: one of the intended technical (deployment diagram), and one of the intended functional architectures illustrating the pilots' data lifecycle (activity diagram).
Rq15	The developed LDT must be able to simulate scenarios and have an advanced capability: Predictive, Prospective or Prescriptive (See Annex 2.2). Specify what advanced capability the developed LDT plans to have.
Rq16	Describe what data, assets, and services will be shared across pilot members, the sectors involved (see Annex 2.6), and the main providers and beneficiaries.
Rq17	Pilots must ensure semantic interoperability of all data exchanged (MIM1)³² within the pilot ecosystem by relying on open and widely recognised standards (e.g., NGSI-LD, LDES, etc.). Applicants must describe the standards used, how common data descriptions are defined or aligned across pilot members, and how semantic interoperability is ensured for data exchange, federation and reuse (See MIM1).
Rq18	Pilots must achieve a score of at least 3 on the LDT4SSC Interoperability self-assessment tool³³ by the end of piloting activities.
Rq19	Indicate which MIMs Plus (MIM0–MIM8)³⁴ the project will engage with (e.g., <i>contribute to, implement, or comply with</i>) and provide a brief justification for this choice —such as existing capabilities, planned developments, or limitations. For each selected MIM Plus, estimate the current and planned level of compliance (e.g., <i>Initial, Partial, or Full</i>)³⁵. The pilot must engage with at least the five foundational MIMs

³² <https://mims.oascities.org/NzWXOO1Fttw4wtqv1Wys/mim1-interlinking-data>

³³ Available soon at <https://ldt4ssc.eu/>

³⁴ <https://living-in.eu/group/7/commitments/mims-plus-version-8-2025>

³⁵ **Initial (1)** – The solution only partially addresses this MIM Plus, with limited features or early-stage implementation. **Partial (2)** – The solution implements key aspects of this MIM Plus but still lacks some elements for full compliance. **Full (3)** – The solution fully complies with this MIM Plus, meeting all major requirements.

	Plus³⁶.
Rq20	Describe whether advanced Digital Technologies such as AI / XR / edge computing are used in the services, how, for what purpose, and how ethical/legal considerations are managed.
Rc2	Pilots are encouraged to use open-source technical components to promote transparency, reduce costs, and foster community collaboration. Any enhancements or customisations developed during the pilot should likewise be shared back with the open-source community to support collective progress and reusable innovation.
Rc3	Pilots are encouraged to pursue alignment with the forthcoming LDT Toolbox as its specifications become available.
Rc4	Pilots are encouraged to federate and interconnect with pilots of WS1 using the SIMPL Data Space Governance Authority ³⁷ .
Rc5	Pilots are encouraged to use open-source technical components to promote transparency, reduce costs, and foster community collaboration. Any enhancements or customisations developed during the pilot should likewise be shared back with the open-source community to support collective progress and reusable innovation.
Rc6	Pilots are encouraged to include in their consortium agreement provisions for the non-exclusive transfer of Intellectual Property or exploitation rights, as well as clear rules for protecting and sharing know-how among consortium members and beyond.
Rc7	Pilots are strongly encouraged to establish, as appropriate to their project scope: • A data catalogue using the W3C Data Catalog Vocabulary (DCAT) for semantic interoperability. • A data management system (e.g., context broker) capable of handling contextualised metadata descriptions (ontologies) such as JSON-LD, RDF, or NGSI-LD for semantic and technical interoperability. • An identity and access management system based on OAuth2, OpenID Connect, or W3C Verifiable Credentials for legal and organisational interoperability. • An ODRL-based data policy to support legal interoperability.

2.6.5 Impact

Rq21	All AI used in public services must comply with the AI Act , particularly if it influences decisions, priorities, resource allocations, or public policies. Pilots must take into account the relevant amendments of the Digital Omnibus on AI, should they come into force³⁸.
Rq22	Explain how the co-created LDTs and services will contribute to governance ,

³⁶ Which are: MIM0 (Accessing Data), MIM2 (Processing Data), MIM1 (Interlinking Data), MIM6 (Securing Data) and MIM3 (Sharing Data).

³⁷ <https://code.europa.eu/simpl/simpl-open/development/agents/governance-authority>

³⁸ See <https://digital-strategy.ec.europa.eu/en/library/digital-omnibus-ai-regulation-proposal>

	efficiency and innovation within and across the pilot members.
Rq23	Describe the anticipated wider socio-economic and environmental effects of your pilot beyond their direct scope, explaining how the eco-design approach contributes to these broader impacts and long-term sustainability. Describe the tangible benefits and long-term impacts expected for the participating of communities, outlining the immediate value proposition, sustainability potential, and measurable indicators to track progress.
Rq24	Pilot's workplan must foresee setting up a contractual framework that guarantees the sustainability of the LDTs and services (legal clauses that guarantee for instance the right to use the data or services, to manage Intellectual Property or the analyses carried out, the post-pilot exploitation rights, etc.)
Rq25	Outline the pilot's strategy for maintaining the LDTs and pilot activities beyond the project's duration, explaining how results will be sustained and scaled up after the pilot phase. Describe how the consortium will elaborate risks & mitigation mechanisms for sustaining the LDT service: • from political, social, technical, operational, business, and legal perspectives, • during the pilot and, e.g., at least 1 year beyond the project. Indicate pilot's initial plans for financial sustainability beyond the project lifetime (e.g. by sourcing new funded projects, fundraising, getting a long-term commitment from the city, etc).
Rq26	Pilots must describe the main expected assets produced in the pilot projects (See Annex 2.9 for a List of potential pilot project assets.)
Rq27	Developed assets must be shared through the LDT4SSC Assets and Services Repository with comprehensive documentation, practical use cases, and detailed implementation guidelines, enabling other users to easily understand, adopt, and reuse these resources in their own LDT projects.
Rq28	Created or adapted data models must be published in open repositories under open licences.
Rq29	The inclusion of proprietary components must be justified and accompanied by a plan demonstrating how an equivalent solution can be achieved with open-source technologies.
Rq30	The developed service must be fully replicable and transferable to other cities and communities across the EU. Describe the measures taken to guarantee Replicability, Transferability and Scalability, e.g. will the solution (re-)use open source components, deployed

	on-premise or in the cloud, and describe how they will be configurable for future use cases. Pilots must describe how the developed assets can be transferred, deployed or adapted in at least one additional local context beyond the pilot sites. Non-public authorities part of the pilot consortium should clearly describe its area of territorial jurisdiction that enables it to replicate the developed LDT based on common needs in other cities, communities and countries of the EU.
Rq31	Pilots are expected to maintain up-to-date documentation of their projects and provide it to the LDT4SSC consortium (at least through Reports and Pilot Deliverables).
Rc8	The use of MIT or Apache open licence is recommended
Rc9	Pilots are advised to record all relevant baseline data , including socio-economic, environmental, and organisational aspects, before starting the project to enable meaningful comparisons in the final Cost-Benefit Analysis .
Rc10	Pilots are advised to assess their level of maturity in eco-design thanks to the General policy framework for the ecodesign of digital services version 2024 ³⁹ at least for the 30 highest-priority criteria at the start of the pilot implementation and then towards the end.

2.7 Ethical and data protection framework

The development and use of LDTs within local and regional authorities are grounded in the principles of transparency and trustworthiness. Consequently, it is essential that their implementation adheres to strong ethical standards.

The ethical practices required from the pilot consortia draw on recognised frameworks and guidelines, including ALTAI (AI HLEG's trustworthy AI assessment list), Fundamental Rights and Algorithm Impact Assessment (FRAIA) and International Organization for Standardization (ISO)/International Electrotechnical Commission (IEC) standards on ethical and societal concerns.⁴⁰

In addition, the LDT4SSC project has established an Ethical Board that participates in the evaluation of CfP proposals and oversees pilot implementations. Ethical aspects are also integrated into the Pilot Support activities through targeted training and mentorship.

³⁹

See https://www.arcep.fr/uploads/tx_gspublication/general_policy_framework_for_the_ecodesign_of_digital_services_version_2024.pdf for more

⁴⁰ Note that pilot consortia must ensure that their contractual obligations under Article 12 (conflict of interest), and Article 14 (ethics) among others are followed. See: Annotated Grant Agreement https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/common/guidance/aga_en.pdf

This Work Strand follows the 'Ethics and Data Protection By Design' approach for the CfP process. It sets several checkpoints with the pilot consortia to raise awareness on ethical and legal challenges and to assess each consortium's capacity to address them effectively. The checkpoints include:

- **Before the submission of application:** In connection to open calls, interested applicants will receive guidance on ethics and data protection during organised sessions. This information will help applicants in identifying ethical and legal challenges and mitigating these upfront. The applicants will also need to complete the **Ethics and Data Protection Self-Assessment** and submit this together with the main application document.
- **During the evaluation of applications:** The Ethical Board will compare ethical self-assessment with proposed work package structures presented in the application (see also Evaluation Process overview).
- **At the start of pilot projects:** Following the announcement of the selected pilots, representatives from the Ethical Board will contact the pilot consortium to discuss ethical needs and provide feedback to mitigate any concerns. For example, they may ask to allocate more time or resources to an ethical aspect that requires attention. This type of feedback should then be accounted for prior to the pilot's start.

The LDT4SSC consortium is responsible for providing overall guidance and monitoring of the project, the CfP procedure and the evaluation. While identifying potential issues falls within the remit of the Ethical Board, the responsibility for the actual work carried out in the pilots, including identifying and mitigating ethical and legal challenges, is beyond the direct control and responsibility of the project consortium.

While it is useful to define common principles, these alone cannot guarantee trustworthy or ethical practices⁴¹. It is therefore recommended (and in some cases required)⁴² that LDT implementations set up mechanisms for oversight, as well as collaborate with a broader LDT community of experts to resolve any ethical issues upfront. Afterall, fair, compliant, and trustworthy access to- control over- and (re)use of- data can only be reached by interdisciplinary efforts and involvement of a broad community of experts and stakeholders.

2.7.1 Compliance with relevant legislation

While ethics go beyond compliance with existing regulation, it is worthwhile to point out that the pilots need to take all required steps to guarantee compliance with the provisions of the relevant EU regulations, inter alia:

- General Data Protection Regulation (GDPR)⁴³
- ePrivacy Directive⁴⁴
- Data Governance Act (DGA)⁴⁵

⁴¹ See e.g., Mittelstadt 2019, <https://doi.org/10.1038/s42256-019-0114-4> >.

⁴² Note that according to the EU's AI Act, data used to build the services and solutions on the platform should be traceable to its origin and should be capable of providing full transparency, if requested.

⁴³ Regulation (EU) 2016/679 (General Data Protection Regulation, GDPR), available at <https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng>

⁴⁴ Directive 2002/58/EC of the European Parliament and of the Council of 12 July 2002, available at <https://eur-lex.europa.eu/eli/dir/2002/58/oj>

⁴⁵ Regulation (EU) 2022/868 of the European Parliament and of the Council of 30 May 2022, available at <https://eur-lex.europa.eu/eli/reg/2022/868/oj>

- Data Act (DA)⁴⁶
- The Interoperable Europe Act⁴⁷, and
- AI Act⁴⁸.

Applicants shall comply with applicable data protection legislation including but not limited to the GDPR and any national implementing laws, regulations, and secondary legislation, in each case as amended, supplemented or replaced from time to time. As a reminder, the GDPR guarantees that the processing of personal data is carried out respecting fundamental rights and freedoms, as well as the dignity of the data subject with particular reference to confidentiality, personal identity, and the right to data protection.

2.7.2 Complaints and appeals

While the LDT4SSC consortium is responsible for evaluating the proposals and in later stages pilots' performance, approved pilot project consortia are responsible for the complaints and appeals stemming from the implementation of their LDTs. We encourage all affected stakeholders to directly contact the pilot consortia for any complaints. If necessary, the LDT4SSC consortium will remain available for support through the info@ldt4ssc.eu helpdesk email address. Complaints regarding ethical issues will be handed over to the Ethical Board for discussion and proposing further action steps.

2.7.3 Confidentiality, privacy notice and usage of applicants' data

The processing of personal data by the LDT4SSC Consortium adheres to GDPR and its Article 5 principles, ensuring lawfulness, correctness, and confidentiality. Personal data will be processed by the LDT4SSC Consortium members and external evaluators responsible for the evaluation process and selection of the pilots. The applicant warrants and represents that, in providing personal data in connection with the proposal, the data subjects have consented to the provision of their personal data and the processing of it by the designated LDT4SSC consortium members, and that the organisation provides the personal data in accordance with applicable law.

Personal data (e.g., email addresses) can also be used for collecting feedback from applicants. Public sharing of data encompasses, among other things, the disclosure of CfP outcome to the LDT4SSC project's web page (<https://ldt4ssc.eu/>). This includes aggregated level statistics such as the number of applicants, applicants per country, the sectors covered. Information on each funded activity, including participant information and abstracts of the activity proposal, may be made available for publication purposes as currently expected:

- Details to be made publicly available **before the end of the pilot**: pilot title, names of the Lead Partner and other beneficiaries, short description (as provided by the applicant in the application template).

⁴⁶ Regulation (EU) 2023/2854 of the European Parliament and of the Council of 13 December 2023, available at <https://eur-lex.europa.eu/eli/reg/2023/2854/oj>

⁴⁷ Regulation (EU) 2024/903 of the European Parliament and of the Council of 11 April 2024, available at <https://eur-lex.europa.eu/eli/reg/2024/903/oj>

⁴⁸ Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024, available at <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>

- Details to be made publicly available after pilot completion: pilot title, names of the Lead Partner and other beneficiaries, awarded funding, updated short pilot description and main pilot results (as provided in the Final Report).

The evaluation and selection of applications will be performed under the appropriate ethical conduct and will respect the confidentiality of the information received⁴⁹. The LDT4SSC consortium will treat any proposal, related information and documents confidentially. Personal data will be processed in accordance with the consortium's Privacy Statement (confirmed by the Ethical Board), or as otherwise indicated throughout the proposal form (e.g., applicants, pilot title, summary description etc.).

The LDT4SSC Consortium members will not be able to discuss in detail or contribute directly to the development of the pilot proposal. Selected Evaluation Committee members evaluating the applications are also obliged to sign the Non-Disclosure Agreement and the Conflict of Interest Statement provided by the Ethical Board.

2.7.4 Dissemination of results and Intellectual Property

By default, LDT4SSC follows the principle of openness to also foster reusability and accelerate research and development of LDTs around Europe, while remaining closed to safeguard privacy and Intellectual Property. Consequently, the default approach of the project is open source also for the piloting results. For specific rights and obligations concerning Intellectual Property Rights of the use and dissemination of results generated by pilot teams through funding obtained via the CfP, the LDT4SSC project will refer to the GA signed with the pilots.

Regarding the dissemination and use of results funded by the CfP, recipients must credit the LDT4SSC project with proper citation, and display the LDT4SSC logo and EU flag.⁵⁰ This includes consistent use of the EU flag and citation throughout the pilot, stating: **"This project has received co-funding from the European Union's Digital Europe Programme under grant agreement No. 101226211"**.



⁴⁹ Note that both the LDT4SSC project consortium as well as the pilot consortia must ensure that their contractual obligations under Article 13 (confidentiality and security) among others are followed. See: Annotated Grant Agreement

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/common/guidance/aga_en.pdf

⁵⁰ See further details from Annotated Grant Agreement under Article 17.2 (visibility) https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/common/guidance/aga_en.pdf and https://commission.europa.eu/system/files/2021-05/eu-emblem-rules_en.pdf

Figure 4. LDT4SSC logo to be displayed by pilots in their communication



**Co-funded by
the European Union**

Figure 5. Consistent use of the EU logo to be displayed by pilots in their communication

3. Call for Pilots Process

3.1 Call for pilots timeline

The open call for pilots supporting the targeted Work Strand 2 - Creating new Local Digital Twins based on common needs is launched based on the following rationale:

- The **Open Call application period** is envisioned to last 2 months from the launch. Call is scheduled to open on 2 February and closing on 2 April 2026.
- After the application deadline, the **review of the pilots** will take two months. This comprises the pre-screening check on eligibility as well as the evaluation by the Evaluation Committee and Ethical Board.
- Following the selection of the winning pilots, the **negotiation period will last one month, followed by another month for necessary preparations** before project launch. This period is meant to draw on the reviews of the Evaluation Committee and Ethical Board, as some recommendations and requests may be provided to the applicants that could imply changes to the pilot in terms of approach, process and consortium. The negotiation period is a timeframe (of 1 month) for providing feedback to applicants and for applicants to address the feedback.
- Each **pilot is envisioned to last between 12 to 18 months from the starting date**.

Table 2 summarises the CfP timeline for Work Strand.

Table 2. Timeline for Call For Pilots for Work Strand 2

Work Strand 2					
	Opening of open call	Closing of open call	Selection of pilots	Start of pilots	Closing of pilots
Open call 2	2 February 2026	2 April 2026	31 July 2026	31 August 2026	31 February 2028

The information on the closing of the pilots in the table, above, is sketched out based on the assumption of a duration of 18 months, although pilots can opt for a duration from 12 to 18 months.

3.2 Documents for submission

As part of the application process, the pilot applicants will be asked to provide the following documentation in order to be considered eligible:

Document type	Description
Application form	General information on consortium details, applicant details, grant allocation request, self-assessment etc.
Financial Form	Details the budget for the pilot proposal.
Letters of Commitment from the Local and Regional Authorities and any other member of the consortium contributing financially to the pilot activities	Statement of intent, affirming the entity's engagement in the LDT4SSC project and its pilot activities.
Letter of intent	Statement of intent provided by another public authority (or service in another public authority if allowed), not in the consortium, supporting that the local challenge addressed in the consortium is also shared in their city or community.
Ethics and Data Protection Self-Assessment	Showing how the applicant will handle people's information responsibly, fairly, and safely, while also respecting individuals' rights.
Ownership and Control Declaration	Certifies that an applicant's organisation is not controlled by entities from ineligible countries.

These documents are referred to in Annex 1 and are published on the project's website⁵¹.

3.3 Where to apply and how

The full application package (written in English) must be submitted via applications@ldt4ssc.eu before the specified deadline (see Chapter 3.1). The application must be filled in using the predefined templates provided on the LDT4SSC project's website.

All the documents must be sent in PDF format, except the Financial Form, which must remain in an Excel format (.xlsx).

The proposal must be submitted by the consortium member acting as the Lead Partner who is also responsible for completing all administrative information and checking all documents. After submitting the documents, a confirmation email will be sent to the applicants verifying that the application was received and stating the time of submission.

⁵¹ <https://ldt4ssc.eu/call-one/>

4. Call for Pilots Evaluation

4.1 Evaluation process

Consortia must submit their applications to the call for pilots under Work Strand 2 by **the 2nd of April 2026 at 23:59 CET**. After this deadline, the review period **will last two months**, including both the pre-screening of the applications and the evaluation by the Evaluation Committee and Ethical Board. After the **ranking and selection** of the pilots, which will be communicated to the applicants, the winning consortia and the selection committee will take one month to finalise contract negotiation and signing, and one additional month to prepare before **project launch**. The process is outlined in the following Figure.



Figure 6. Expected piloting process for all five calls

The following sections outline the process for each step:

Pre-screening of the applications through eligibility check

After the deadline for the call has passed, each application will be pre-screened for an eligibility check as a first step. This will ensure that only relevant applications are considered for the evaluation of the Evaluation Committee and Ethical Board (see below).

Applications will first be anonymised and then assessed by the eligibility check grid. The criteria are listed in the table below:

Table 3. Eligibility Check Grid for Work Strand 2 (legal, financial, technical, ethical)

	Eligibility criterion	How it's demonstrated	How it's assessed
General	All required documents and ownership and control declarations have been submitted in the right format (PDF & Excel) and on time. The application files are submitted in English.	Application Form, Financial Form, Letters of Commitment, Letter(s) of Intent, Ethics and Data Protection Self-Assessment	pass/fail
	The proposal has been submitted using the Open Call template. This includes respecting the page limit for the proposal template.	Application Form, Financial Form, Letters of Commitment	pass/fail
	All questions have been answered (within the character limits).	Application Form, Pilot Budget, Ethics and Data Protection Self-Assessment	pass/fail

	Eligibility criterion	How it's demonstrated	How it's assessed
Legal	Pilot consortium consists of at least two local or regional public administrations from two different eligible countries, together with at least one additional partner drawn from one of the following categories: • Private entity (e.g., service provider) • Private association (legal status) • Trusted third party • Private representative of a use-case sector	Application Form, Letters of Commitment	pass/fail
Financial	Pilot consortium covers at least 50% of the pilot budget as co-financing.	Applicants demonstrate their capacity to provide at least 50% co-financing in an added budget table and with the Letters of Commitment	pass/fail
	Support requested is not higher than 1.000.000 euros per consortium.	Financial Form	pass/fail
	Support requested is not more than 500.000 euros per third-party across pilots.	Financial Form	pass/fail
Project Management	The project description and workplan should adopt the LDT4SSC methodology, following the 4 phases: Explore, Validate, Define and Implement. Applicants are expected to submit a workplan revolving around the LDT4SSC pre-defined Pilot Deliverables (PD).	Application Form	pass/fail
Use Case	Pilots must include at least one cross-sectoral innovative and citizen-focused use case with two services.	Application Form	pass/fail
	Public authorities part of the pilot consortium must share a common need and address a shared local challenge.		

	Eligibility criterion	How it's demonstrated	How it's assessed
Technical Implementation	Pilots must develop a new common LDT based on their common need. Each public authority participating in the pilot must implement its own instance of the commonly developed LDT. Hence, there must be at least 2 instances of the common LDT, one for each public authority, which allows executing the two services related to the identified cross-sectoral use case.	Application form	pass/fail
	The developed LDT must be able to simulate scenarios and have an advanced capability: Predictive, Prospective or Prescriptive.	Application form	pass/fail
Ethical	Ethics and Data Protection Self-Assessment completed and submitted.	Application Form, Ethics and Data Protection Self-Assessment	pass/fail

This eligibility check process should take no longer than 4 weeks. Applicants are encouraged to submit their applications as early as possible before the deadline.

Evaluation by Evaluation Committee and Ethical Board

Once applications have successfully passed the eligibility check, an **Evaluation Team** will assign **expert evaluators** to an **Evaluation Committee** based on their respective areas of expertise.

Each application will be reviewed by two external evaluators and one internal evaluator. All three evaluators will assign scores following a structured evaluation framework which is fully aligned with European priorities and upcoming policies, the Digital Europe Programme (DEP), the Mission on Climate-Neutral and Smart Cities, as well as the DS4SSCC-DEP. These will be then combined to form an aggregate score, which will determine the application's position in the final ranking. The assessment will focus on three primary dimensions: **Excellence, Impact, and Quality and Efficiency of the implementation, as shown in Table 4.**

Table 4. Evaluation criteria

	Evaluation criterion	How it's demonstrated	Minimum pass score	Maximum pass score
1	Excellence	1.1 The proposal clearly defines its objectives and demonstrates strong alignment with EU priorities and LDT4SSC challenges. 1.2 The use cases are well-defined, feasible, and clearly demonstrate the added value of interconnecting LDTs and sharing services. 1.3 The technical and functional architectures are coherent, complete, and supported by appropriate standards, tools, and diagrams. 1.4 The data governance approach is robust, and ethically compliant with relevant regulations (see Section 2.9.1) (and demonstrates responsible integration of AI/XR technologies, when applicable). 1.5 The methodological approach is well-structured, and clearly describes the data lifecycle.	6	10
2	Impact	2.1 The pilot demonstrates clear and measurable benefits for local communities, including socio-economic and environmental gains. 2.2 The proposal shows strong potential for replication across EU cities and meaningful contribution to EU LDT ecosystems and standards. 2.3 The outputs are clearly defined, of high-value, and committed to open publication of data models and services where feasible. 2.4 The pilot includes a credible plan for involving relevant stakeholders and end-users throughout the project lifecycle. 2.5 The proposal presents a realistic and well-supported plan for sustaining and scaling the pilot after project completion.	6	105

	Evaluation criterion	How it's demonstrated	Minimum pass score	Maximum pass score
3	Quality and efficiency of implementation	3.1 Existing community platform, digital twin, or emerging data space is sufficiently mature to reach described objectives 3.2 The consortium demonstrates strong expertise, well-defined roles, and adequate political and organisational support. 3.3 The work plan is clear, realistic, and allocates resources efficiently across tasks and partners. 3.4 The proposal provides a feasible plan for managing data, identity, access, and policies across platforms. 3.5 Risks are clearly identified with credible mitigation measures, and eco-design principles are meaningfully integrated.	6	10

It is envisaged that the **Evaluation Committee** and **Ethical Board** will work in parallel to review the applications during the same timeframe. The Ethical Board's assessment will focus on the pilots' Ethics and Data Protection Self-Assessment and considerations regarding ethical implications of the application's aspects. These include evaluating issues related to data governance, privacy, consent, and transparency in the collection, processing, and sharing of local data. The Ethical Board will also assess potential biases in data modeling and simulation, fairness in the representation of communities, and the risk of reinforcing social or spatial inequalities through digital replication. These assessments will be based on a review of the ethics and data form, as well as a review of the section on "Alignment with Ethical Principles" on the Application Form.

The evaluation will take place in two stages:

Stage 1. Individual Evaluation

In the first instance, evaluators will assess and score the applications individually, providing written justifications in line with a standard template. Alongside this process, selected members of the Ethical Board will review the sections that are relevant for the ethics and the accompanying Ethics and Data Protection Self-Assessment submitted by the applicants. This draws on the evaluation framework and the evaluation criteria outlined above.

Stage 2. Consensus Group Meeting

Following the completion of individual evaluations, respective members of the given Evaluation Committee for each Work Strand and relevant Ethical Board members who reviewed the applications will convene to discuss their assessments and compare findings. The aim is to reach a consensus on comments and scores, ensuring consistency across evaluations. A moderator from the Evaluation Team will facilitate the discussion, guiding the group towards agreement and guaranteeing that the proposals are evaluated fairly and in accordance with the established criteria.

The outcome of this process will be a consolidated evaluation report, including score justifications and, where applicable, any dissenting opinions. The report is formally signed by the evaluating experts, the Ethical Board representatives involved, and the moderator. The

table below visualises the template used by evaluators to assign individual scoring before drawing an aggregate average from three scores.

Table 5. Preliminary ranking by Evaluation Committee for one pilot

No.	Evaluation criteria	Content evaluated	Score
1	Excellence		Out of 10
2	Impact		Out of 10
3	Quality and efficiency		Out of 10
			Total out of 30

Final decision by the Steering Committee

A Steering Committee shall consist of one representative from each private entity from the LDT4SSC consortium. It is responsible for making decisions concerning the awarding of pilots in collaborations with the Ethical Board and Evaluation Committee.

While the final score is determined by the ranking of the evaluation criteria, it is expected that the Steering Committee will make the final decision based on the following considerations:

- **Priority of the evaluation criteria:** Proposals with an equal final score may be ranked according to the scores they were given for criterion 'Excellence'. When these scores are equal, priority will be based on their scores for the criterion 'Impact'.
- **Recommendations from the Evaluation Committee:** Specific feedback about the scope and quality of the proposed pilots should be taken into account when deciding on the final ranking.
- **Geographical balance:** When possible, geographical diversity should be prioritised in order to ensure that the final set of shortlisted pilots will include different areas of Europe. In the following call rounds, also previously awarded pilots' geographical background should be considered.
- **Domain-based balance:** When possible, diversity of domains should be prioritised in order to ensure that the final set of shortlisted pilots will cover a variety of domains. In the following call rounds, also previously awarded pilots' domains should be considered.
- **Varying maturity levels:** While all applicants must meet sufficient maturity levels for digitalisation, it should be respected that applicants with lower maturity would get a chance to participate in the piloting programme, if deemed reasonable.

Decision communication

The decision of the winning pilots will be communicated shortly after the two-month review period. Applicants will be contacted via email with the final decision and score.

Contract negotiation and signing

Drawing on the announcement about the selected pilots, the contract negotiation period starts, which is expected to last one month before project launch. This period draws on the

reviews of the Evaluation Committee and Ethical Board, as some recommendations and requests may be provided to the applicants that could imply changes to the pilot in terms of approach, process and/or consortium. The negotiation period is thereby a timeframe for providing feedback to applicants and for applicants to address the feedback. It encompasses the process of negotiating, providing and checking the additional information (such as the Ownership Control Declaration (OCD), if necessary) and signing the contract.

Final preparations

Applicants will still have one month after signing and before the start of the project to finalise preparations, such as checking for capacity and resources and ensuring that they are ready in time for project launch.

Project start

Once the contract negotiation and signing is done and preparations are in place, the pilot project kicks off.

4.2 Checklist

To ensure that all necessary steps for successful application submission have been followed, applicants are recommended to run through the following pre-submission checklist (Table 7).

Table 6. Checklist for applicants before Submission

	Activity	Yes/No
#1	You have prepared all documents for the submission: Application Form, Financial Form, Letters of Commitment, Ethics and Data Protection Self-Assessment, Ownership Control Declaration	
#2	Your documents are in the right format (PDF & Excel) and answer within the character limits. All sentences and images exceeding the limitations will be hidden from the application for evaluation.	
#3	Your consortium has at least two local or regional public administrations from two eligible countries on board.	
#4	All partners of the pilots are from the eligible countries, and the activity of the pilot will take place in the eligible countries.	
#5	You have described at least two use cases across priority sectors in the Application Form.	
#6	Your consortium does not request more than 1,000,000 Euros and will cover at least 50% of the pilot budget.	

#7	Your pilot complies with the minimal technical and non-technical requirements as described in the technical framework.	
#8	Your LDT pilot complies with relevant regulations, such as GDPR, DGA etc.	

5. Applicants support

5.1 Support during the application phase

The project provides a comprehensive suite of resources to support applicants to the CfP.

- The **Project Website**⁵² serves as the central point of reference with information provided on the following:
 - **Frequently Asked Questions (FAQ) Section** – the FAQ addresses frequently asked questions and helps applicants avoid common errors during proposal preparation.
 - **Info sessions dedicated to the call** – 1-2 information sessions will be organised to provide additional information and answer questions from applicants, one organised around the call opening, and one halfway throughout the application phase.
 - **Matchmaking sessions** - 1 session will be organised to promote contact and networking between potential interested applicants.
 - **Helpdesk** – applicants can submit their questions via the email address info@ldt4ssc.eu.
- The **LDT4SSC Knowledge Hub**⁵³ provides guidance, resources, and examples from previously awarded pilots, supporting both the preparation and implementation of new proposals. It also collects and organises the project's knowledge outputs, providing applicants and stakeholders with easy access to strategies, tools, and guidance.
- Available **EC policy documents** including priority areas such as the Clean Industrial Deal⁵⁴ and New European Bauhaus⁵⁵ initiative to define at least two relevant use sectors central in your pilot proposal.
- A **matchmaking platform**⁵⁶ is launched to facilitate connections between stakeholders.

⁵² See <https://ldt4ssc.eu/>

⁵³ See <https://knowledgehub.ldt4ssc.eu/>

⁵⁴ https://commission.europa.eu/topics/eu-competitiveness/clean-industrial-deal_en

⁵⁵ https://new-european-bauhaus.europa.eu/index_en

⁵⁶ See <https://www.b2match.com/e/local-digital-twins-smart-communities>

5.1.1 The LDT4SSC Pilot Helpdesk

The **LDT4SSC Pilot Helpdesk** (operated via the email info@ldt4ssc.eu) acts as the single entry to ensure clear, timely and accountable support to both applicants and awarded pilot consortia from application to implementation.

The Pilot Helpdesk provides **practical operational support to applicants on administrative, procedural, ethical/data-protection, and technical matters** relevant to the pilot application process. It also offers implementation support for example regarding reporting requirements, use of project templates, and access to mentoring opportunities.

The Helpdesk will **not provide legal representation, financial audit services, or substitute formal legal or accounting advice.**

All questions and answers will be published on a dedicated Pilot Support page, which will include a comprehensive FAQ section and relevant contact information.

5.1.2 Confidentiality and data protection

All personal data and case information processed by the Helpdesk will be handled in compliance with GDPR and the project Privacy Statement. Information shared with external evaluators, partners or third parties will be limited to what is strictly necessary and only transmitted after appropriate confidentiality safeguards are in place.

6. Training and supporting activities

Each pilot team is recommended to designate participant(s) to take part in training and support activities. These sessions are designed to address the specific needs of decision-makers and IT experts from local authorities, providing tailored content aligned with their roles, expertise, and knowledge levels.

For each pilot, the expected effort allocation across different roles involved in support activities should be estimated along with the corresponding budget commitment. A suggested estimated contribution is of 80-100 hours across the duration of the pilot. More specific details about the estimated effort required will be communicated closer to the beginning of the pilot launch. Training and support activities may include workshops, process support, mentoring sessions, and periodic evaluation meetings.

7. Impact and expected outputs

In the application form (Section 4.1), prospective pilot consortia are expected to describe the tangible changes their initiative is expected to deliver for participating communities, as well as its longer-term impact. This includes outlining anticipated socio-economic and environmental effects that may extend beyond the pilot's direct scope. This information helps to ensure that the pilots selected are aligned with the expected impact of the call.

Expected outputs and KPIs (application phase)

Applicants should provide information on the pilot's **context** and **inputs**, and specify a set of **output indicators** supported by relevant **Key Performance Indicators (KPIs)**. In particular, pilots are expected to:

- describe the main **expected outputs/assets** to be produced (e.g. services, tools, data models, datasets, methodologies, or other reusable assets);
- define a coherent set of **KPIs** to measure progress and achievement of these outputs; and
- explain how KPIs will be monitored throughout the pilot lifetime.

Outcomes and longer-term impact

In addition to outputs and KPIs, applicants should identify the **desired outcomes** that the pilot aims to contribute to, and explain how these outcomes support the intended **longer-term impact**. Where feasible and appropriate, local administrations are encouraged to select measurable short-term outcomes that can be reasonably linked to pilot outputs.

It should be noted that the learning and results generated by earlier pilots (e.g. services developed, tools tested, and experience gained) may become inputs for later pilots.

As illustrated in Figure 7 below, therefore:

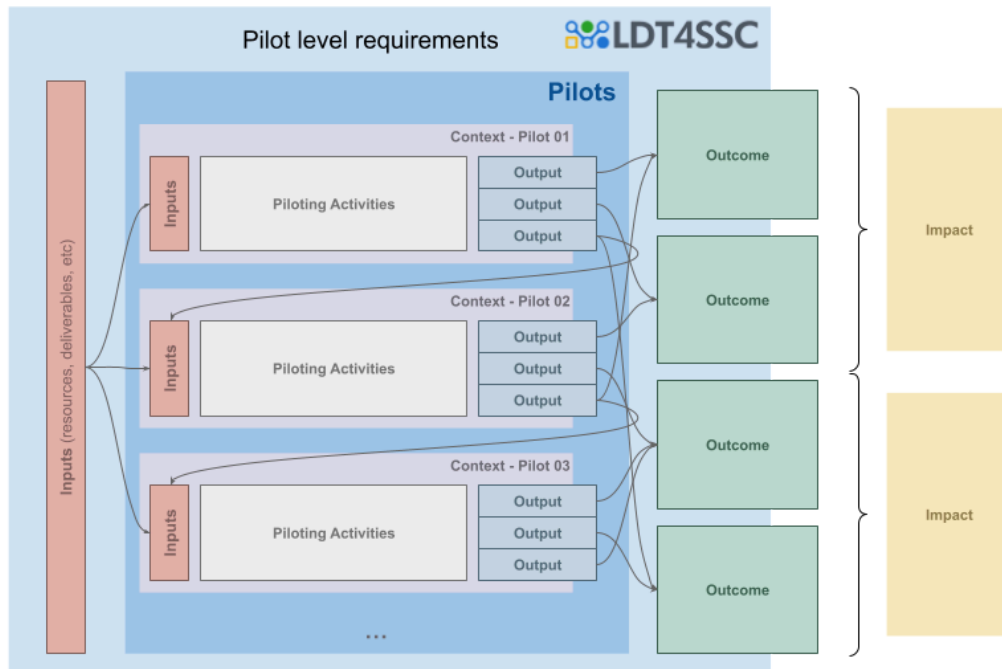


Figure 7. Pilot level requirements for impact assessment

LDT4SSC Impact Assessment Framework

As illustrated in **Figure 8**, the LDT4SSC Impact Assessment Framework structures the pilot intervention logic as follows:

- **Inputs:** resources, enabling conditions, and contributions supporting the pilot, originating both from the project and from the pilot's local context.
- **Activities:** actions implemented by the pilot consortium at pilot level.
- **Outputs:** direct deliverables produced by the pilot (the main assets and measurable results).
- **Outcomes:** broader changes that pilots aim to enable, linked to project-level objectives.
- **Impacts:** longer-term effects aligned with wider EU priorities, including the EU Green Deal and the UN Sustainable Development Goals.

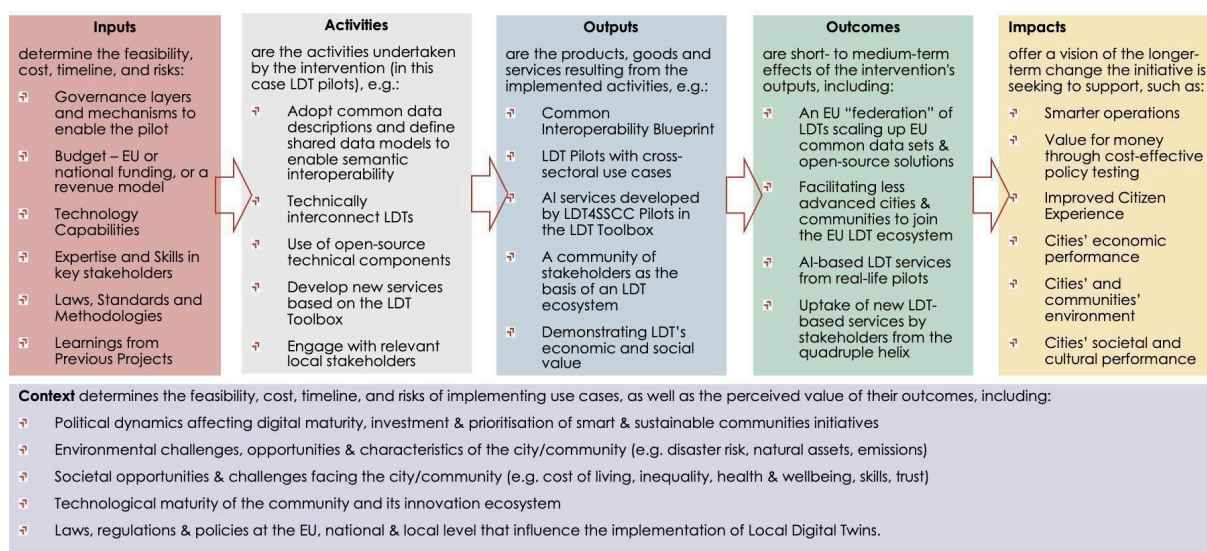


Figure 8. LDT4SSC Impact Assessment framework

Data collection and reporting

The pilot coordinator will submit reporting deliverables on behalf of the consortium. Pilots will be required to submit:

- an **interim report** (mid-term), and
- a **final report** (end of implementation).

Reporting will include a self-assessment component, supported by a questionnaire designed to capture baseline values for selected indicators at pilot kick-off.

In both interim and final reporting, pilots are expected to provide:

- an overview of activities implemented by consortium members
- a summary of resource use (as relevant and proportionate)
- progress updates against the KPIs defined in the application and agreed with the LDT4SSC Consortium.

Data collection approach

The use of ad-hoc surveys, questionnaires, and other data collection tools will be assessed and, where appropriate. Evidence collected through monitoring of pilot activities, outputs and outcomes will support the assessment of pilot performance and inform the interpretation of longer-term impact.

Annex 1 - Required Documents

All documents listed below can be found and downloaded from the LDT4SSC website and should be emailed as an application package to applications@ldt4ssc.eu.

Annex 1.1 Financial form

Annex 1.2 Commitment letter

Annex 1.3 Ethics and data protection self-assessment template

Annex 1.4 Ownership Control Declaration

Annex 1.5 Letter of Intent

Annex 2 - Information for context

Information for context: in support of the requirements and recommendations additional information is provided in the following annexes for context purposes.

Annex 2.1 LORDIMAS digital maturity assessment

The **LORDIMAS digital maturity self-assessment**⁵⁷ is available for local, metropolitan, and regional governments. This tool focuses on multiple aspects of wider EU-led policy initiatives which are related to governance, service design, data management, interoperability, service delivery, technology and networking.

While filling in the LORDIMAS self-assessment, governments can provide their inputs for the assessment and visualise their results on an interactive dashboard in real time. In addition, the tool enables its users to:

- track their transformation progress over time,
- benchmark themselves against others,
- share best practices,
- get tailored policy recommendations.

While it is **optional** for pilots to fill the assessment for each local or regional government of the consortium, the score may complement the answers to the questions related to digital maturity in the application form.

LORDIMAS
LIVING-IN.EU

The 6 Stages of Digital Maturity

Where are you?

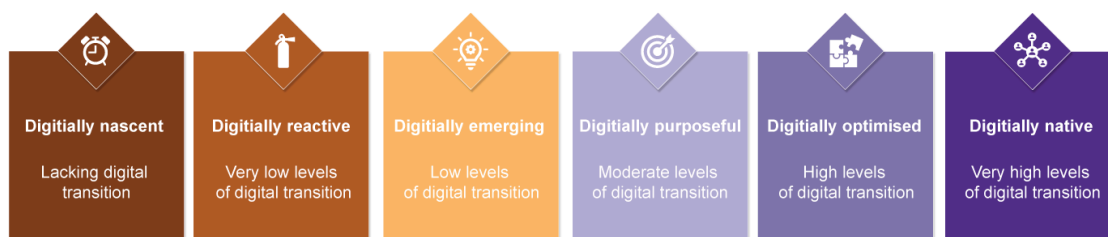


Figure 9. LORDIMAS 6 stages of Digital Maturity

⁵⁷ See [MaturityAssessment-help | LORDIMAS EN](#)

Annex 2.2 Capabilities of a Local Digital Twin

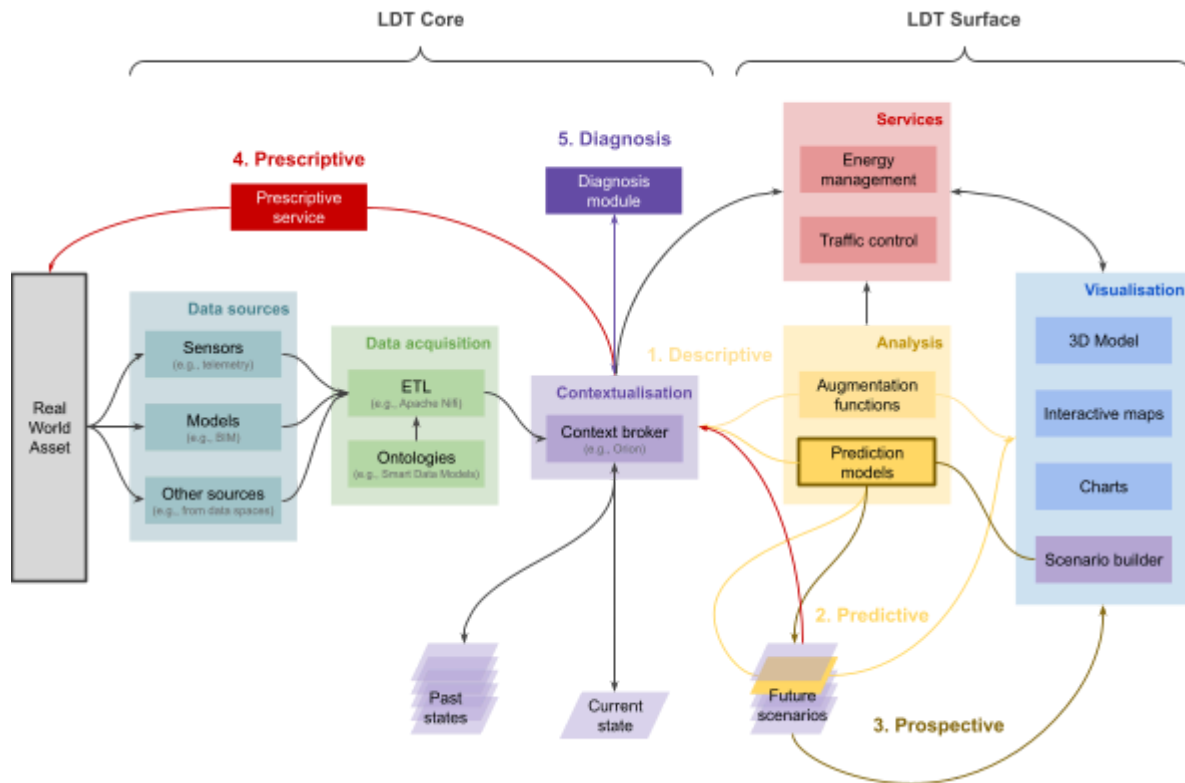


Figure 10. Capabilities of a Local Digital Twin.

The Context Information Management (CIM) European Telecommunications Standards Institute (ETSI) Industry Specification Group (ISG) presents several capabilities for digital twins⁵⁸.

1. **Descriptive:** Informs on the current state of the real-world assets. The Descriptive Twin presents past and current values of some of the real-world asset characteristics. These characteristics can be static (e.g. a building geometry) or dynamic (e.g. sensor measurements). The connection between the real-world asset and the Digital Twin is bidirectional: a change made in the Descriptive Twin is reflected (in real-time or not) onto the real world asset (actuation). The Descriptive Twin capability does not restrict to metrics collected from the real-world asset and can be augmented with computed functions (e.g. distance to a fixed point of a moving real world asset; deviation from normal of a room temperature)..
2. **Predictive:** Extends the descriptive twin capability by providing predictions on the way the real asset could evolve in the future, using predictive models to envision future scenarios.
3. **Prospective:** Conducts "what-if" analyses to evaluate the potential consequences of actions, extending the predictive model to multiple possible future scenarios..

⁵⁸ See section 6.2:

https://www.etsi.org/deliver/etsi_gr/CIM/001_099/017/01.01.01_60/gr_CIM017v010101p.pdf

4. **Prescriptive:** Extends (or, in extreme cases, executes) the prospective capability with suggested actions on the real system to achieve a given objective based on the analysis.
5. **Diagnostic:** Explains situations or alerts about deviations from expected conditions. Capability for evaluating what happened, especially in the case of a malfunction of the real asset.

Annex 2.3 European Green Deal Policy areas, New European Bauhaus initiative values and LDT4SSC key challenges

European Green Deal Policy areas
• Clean energy, • Sustainable industry, • Building and renovation, • Farm to fork, • Eliminating pollution, • Sustainable mobility, • Biodiversity and ecosystem health and • Sustainable finance.
New European Bauhaus initiative values
• Sustainability, from climate goals to circularity, zero pollution and biodiversity; • Inclusion, from valuing diversity to securing accessibility and affordability; • Aesthetics and quality of experience for people, through design and cultural benefits.
LDT4SSC key challenges
• Climate change, • Energy efficiency, • Mobility, • Waste management, • Biodiversity preservation, • Air quality and • Water management.

Annex 2.4 Example of a (twice!) cross-sectorial use case

A region develops an LDT to improve the management of their road infrastructures by combining data from different sectors. The LDT brings together information about the roads (e.g. their width, condition and maintenance status) with dynamic environmental data, including weather forecasts and flood risk metrics.

This combination enables the city to provide valuable digital services that cross traditional-sector boundaries. For instance:

- Road engineers can use the LDT to identify which areas require urgent repairs, especially when poor road conditions overlap with upcoming storms or flood risks. This allows maintenance efforts to be better planned and more efficient, reducing long-term damage and improving public safety.
- At the same time, the LDT provides a second data service that is useful to special convoys, heavy vehicles and emergency services as it helps in identifying the safest and most reliable roads in real time. This is particularly important during severe weather, when certain routes may become dangerous or impassable.

By connecting infrastructure and environmental data in one shared data platform, the LDT supports smarter planning, quicker response times and more resilient public services. It illustrates a clear example of how cross-sectoral data integration can lead to more informed decisions and improved outcomes for communities in multiple sectors.

Annex 2.5 Data governance

Data Governance to...

- Manage the interconnection through use cases (interconnection of LDTs, etc.);
- Capitalise on all use cases;
- Involve all stakeholders for each use case, from the ideation phase to the industrialisation phase of services/digital twins, whether or not they decide to integrate AI.

The implementation of data governance requires:

- a set of common practices,
- standard processes, and
- standard contracts underlying each use case deployment.

For the creation of digital ‘Commons’ (e.g. data spaces, jointly-created digital twins, etc.), data governance also provides a framework and regulations.

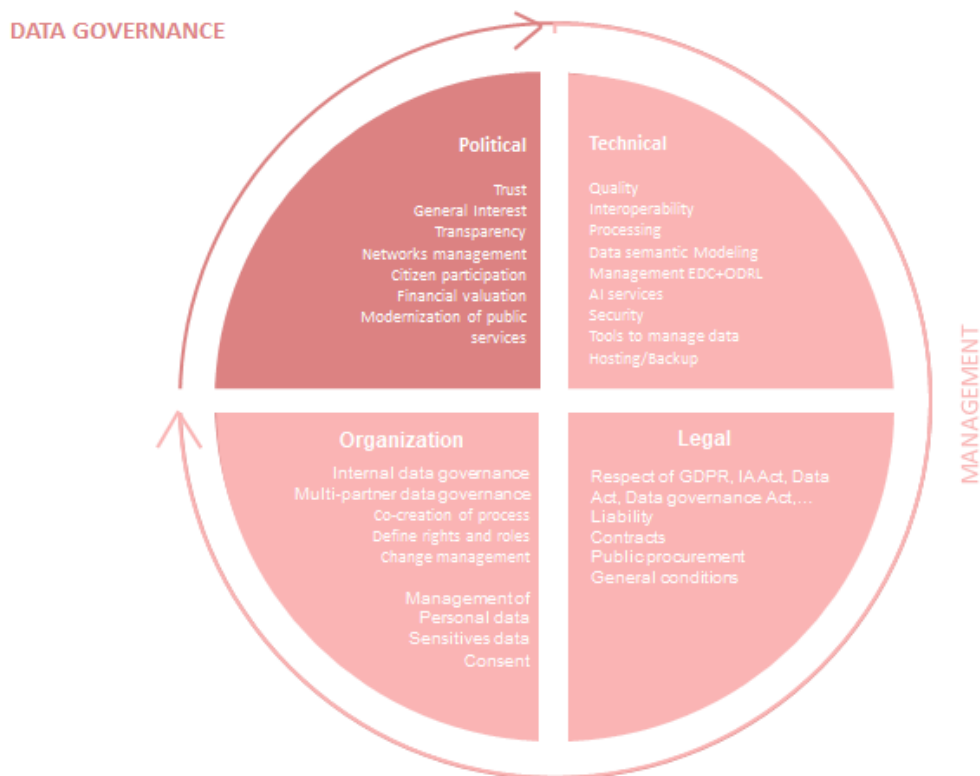


Figure 11. Data governance – A Pilot's strategy throughout political, technical, legal and organisational aspects

Figure 11 illustrates the scope of Data Governance across four dimensions:

- **Political:** this dimension covers the strategic aspects related to data governance.

- **Technical:** this dimension, addressed in terms of technological choices in the technical section of calls for projects, must be linked to other aspects of data governance.
It will deal with topics like how pilots will coordinate with various stakeholders, ensure the technical maintenance and development of the digital twin, etc. It will also aim at understanding the technical infrastructure and architecture.
- **Legal:** this dimension aims to pinpoint the several compliance and legal aspects related to governance, such as how the terms and conditions of use are defined (as previously mentioned), but also it's about broader sections like compliance with EU regulations such as the GDPR etc. It highlights a clear issue of data governance that feeds into the services created (Digital Twins) and of contractualisation between the pilot members at the end of the pilot.
- **Organisational:** this dimension deals with “traditional aspects” one might consider in the scope of data governance, i.e. the management teams and several stakeholders involved, the internal processes etc.

NB: At every stage of the pilot, project-relevant questions should be raised, appropriate clauses included, and recommendations implemented. Key guidance for each stage is provided in D3.4 Non-Technical Resources for Pilots.

NB: When applying AI technologies, particular attention must be given to data governance to ensure full compliance with the AI Act.

Annex 2.6: NACE (“statistical classification of economic activities” in the European Community) Rev. 2 – Level 1 (Sections)

A – Agriculture, forestry and fishing

B – Mining and quarrying

C – Manufacturing

D – Electricity, gas, steam and air conditioning supply

E – Water supply; sewerage, waste management and remediation activities

F – Construction

G – Wholesale and retail trade; repair of motor vehicles and motorcycles

H – Transportation and storage

I – Accommodation and food service activities

J – Information and communication

K – Financial and insurance activities

L – Real estate activities

M – Professional, scientific and technical activities

N – Administrative and support service activities

O – Public administration and defence; compulsory social security

P – Education

Q – Human health and social work activities

R – Arts, entertainment and recreation

S – Other service activities

T – Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use

U – Activities of extraterritorial organisations and bodies

Annex 2.7: High-Level architecture Overview of an LDT with the EU LDT Toolbox⁵⁹

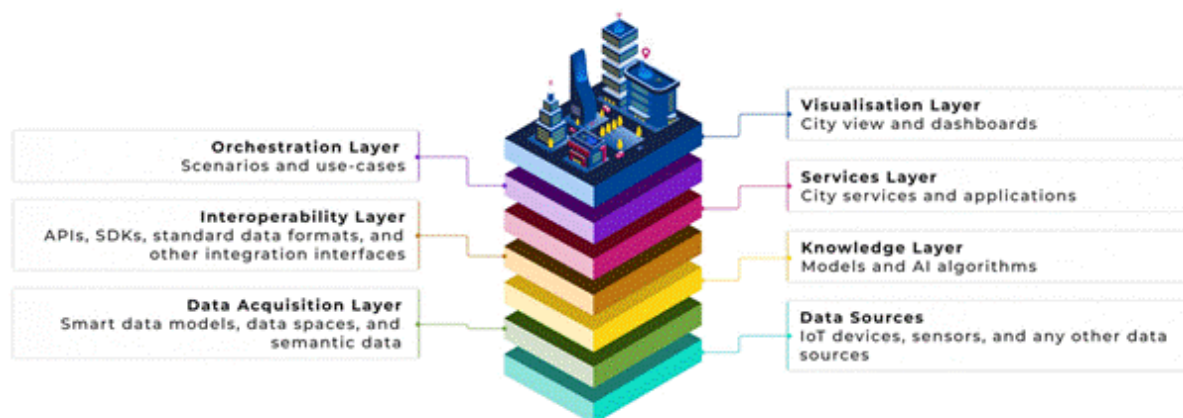


Figure 12. High-Level architecture Overview of an LDT with the EU LDT Toolbox.

In the EU LDT-Toolbox, a Local Digital Twin is described as a system-of-systems typically structured around seven common architectural layers:

1. **Data Sources layer.** The “inputs” layer: IoT devices, sensors, databases, open APIs, and data spaces that provide both real-time and static data from the urban environment.
2. **Data Acquisition layer.** This layer is responsible for bringing data from heterogeneous sources into the LDT ecosystem and applying the first processing steps—such as ingestion, validation/cleaning, transformation and harmonisation—so that the resulting datasets are consistent and ready for downstream analysis and reuse. In the EU LDT-Toolbox specifications, it corresponds closely to workflow stages such as data ingestion and data transformation. Typical toolbox support for implementing this layer includes the EU LDT Data Platform (data consumption and context brokering capabilities) and EU LDT Data Space Ready (connectivity and integration with data spaces).
3. **Knowledge layer.** This layer builds on harmonised data to generate actionable knowledge through advanced analytics, algorithms and AI/ML. It typically covers activities such as data exploration and feature engineering, development and training of models, inference and production of derived datasets, and—where relevant—synthetic data generation to support testing, model robustness or privacy-preserving use cases. Within the EU LDT Toolbox, this layer can be supported by tools such as the EU LDT AI Notebook (for experimentation and model development), the EU LDT Data Modeller (for structuring and managing data models), and EU LDT Federated Learning (for distributed model training across multiple data holders).
4. **Interoperability layer (orthogonal).** This is a cross-cutting, “across-the-stack” layer that enables consistent communication and coordination between the subsystems composing an LDT as a system of systems. It ensures that components can interoperate through agreed interfaces and semantics, and it supports long-term extensibility by

⁵⁹ See <https://interoperable-europe.ec.europa.eu/collection/ldttoolbox/knowledge-center>

allowing the LDT to integrate external systems and services that were not originally foreseen. In this respect, the EU LDT Toolbox treats interoperability as a central design concern, with Minimum Interoperability Mechanisms (MIMs) and relevant standards acting as the “glue” that enables compliant, reusable, and scalable interactions across components.

5. **Services layer.** This layer provides the functional capabilities consumed by LDT use cases and interfaces with specialised municipal services and operational applications (e.g., energy, water, mobility/traffic, environment). It exposes domain and platform functions through well-defined interfaces (typically APIs and/or event streams) so they can be reused and composed across scenarios. In the building-block view, service-like components include, for example, a Context Broker, which integrates and manages context information from multiple systems and can act as an integration gateway—potentially also supporting actuation pathways—alongside related data-service components that enable access, querying and dissemination of LDT data.
6. **Orchestration layer.** This layer composes and coordinates data models, services and workflows to implement end-to-end LDT use cases, including the execution of what-if analyses and simulations. It provides the control plane for managing scenarios and ensuring that the right components are invoked in the right sequence (or in parallel) with the appropriate data and parameters. In the EU LDT Toolbox, this corresponds to capabilities such as use case/scenario management and data workflow & component orchestration, including task scheduling, dependency management, parallel execution, operational monitoring and logging.
7. **Visualisation layer.** This layer delivers the user-facing interfaces that enable stakeholders to explore, understand and communicate LDT information and results through advanced visualisation and interaction mechanisms (e.g., dashboards, 2D/3D views and digital maps). In the EU LDT Toolbox specifications, this includes capabilities such as geospatial visualisation, analytical visualisation components for interpreting model outputs and indicators, and XR services (AR/VR) that support richer interaction, engagement and immersive exploration where appropriate.

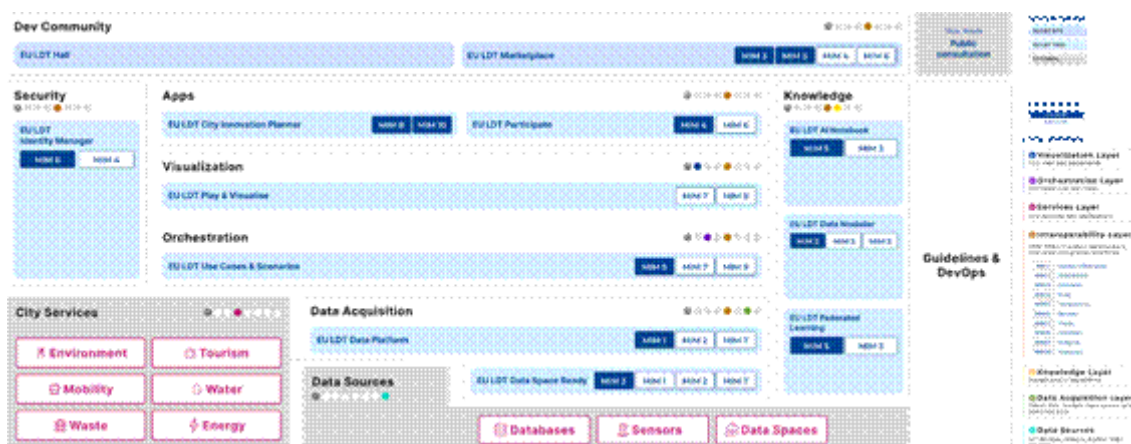


Figure 13. LDT Toolbox solutions structure with layers and MIMs Plus

Annex 2.8: LDT4SSC Methodology⁶⁰

The LDT4SSC methodology is structured in four main phases:

1. **EXPLORE - Ideation:** generating user-centred use cases.
2. **VALIDATE - Specifications:** specifying the functionalities of the solution that meets the need.
3. **DEFINE - Prototyping:** testing the first solutions quickly and cost-effectively.
4. **IMPLEMENT - Deployment:** structuring an operational, scalable and sustainable project.

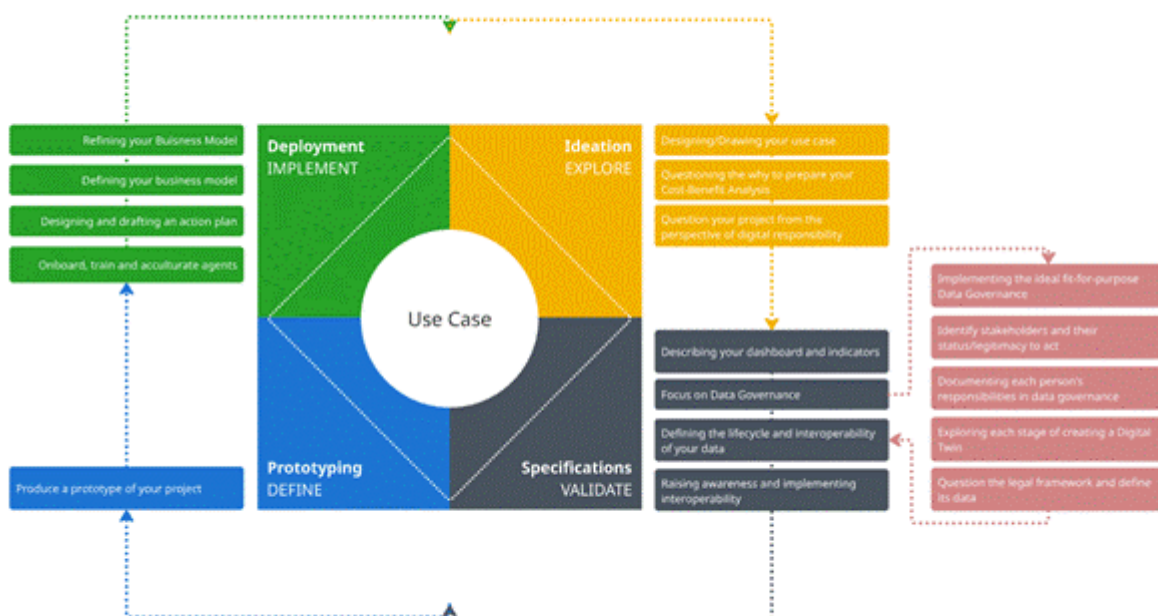


Figure 14. Overview of the LDT4SSC Methodology steps and proposed workshops.

Each of these stages mobilises appropriate tools and resources. The methodology is distinguished by the particular attention paid to **data interoperability**, **contextualisation** (semantic modelling) and the use of an infrastructure that **facilitates re-use**.

It is a method based on the two editions of the French CAPACities programme.

⁶⁰See

https://knowledgehub.ldt4ssc.eu/resources_content/non_tech_resources/#the-ldt4ssc-methodology

Annex 2.9: List of pilots projects produced assets

In this context, an asset is any digital artefact produced or used by a pilot that enables the Local Digital Twin to operate, be reused or shared. For instance:

- **Datasets** : raw, cleaned or contextualised datasets used by the LDT.
- **Data models / ontologies**: contextual models and schemas.
- **Algorithms and code**: processing, transformation, analytics code, scripts, and software components.
- **AI models and ML artefacts**: trained models, model artefacts and metadata.
- **Assets for Marketplace publication**: deployable KServe packages, storage URIs, inference API specs (V2-style).
- **Services / microservices**: deployable service components.
- **Visualisations and UX components**: dashboards recipes, 2D/3D views, XR/VR components.
- **Simulations / scenario builders**: models and workflows for predictive/prospective/prescriptive analyses.
- **Deployment artifacts / orchestration configs**: Docker/Kubernetes manifests, KServe InferenceService YAMLS, deployment scripts.
- **Documentation, implementation guidelines and metadata**: user guides, API docs, data catalog entries (e.g., DCAT), provenance and governance info.
- **Contracts / governance templates**: legal/contractual clauses, data policy templates (e.g., ODRL-based).
- **Pilot deliverables / reports**.

This is a non-exhaustive list.

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